ALMOST-EQUIDISTANT SETS AND TRIANGLE-FREE GRAPHS

HYUNJUN YI, BRYAN BU, KARAM GILL

Abstract. A set of points in $\mathbb{R}^d$ is almost-equidistant if among any three of them some two are at distance 1. Let $f(d)$ be the largest size of an almost-equidistant set in $\mathbb{R}^d$; the conjecture $f(d) = O(d)$ is open and the best known upper bound is $O(d^{4/3})$ (Kupavskii, Mustafa, Swanepoel). We study f through the triangle-free graph formed by the pairs at non-unit distance.

Our main result is a bound with the sharp leading constant for sets without short distances: an almost-equidistant set in $\mathbb{R}^d$ in which no two points are at distance less than 1, in particular every two-distance almost-equidistant set whose second distance exceeds 1, has at most $(d+1)^{4/3}+6(d+1)$ points and at most $(d+1)^{4/3} + 2(d+1) + 100(d+1)^{2/3}$ points; the same bounds then hold for all two-distance almost-equidistant sets. The new ingredient is the positivity of the third moment $\sum_{i,j}\langle v_i, v_j\rangle^3$, which for sets built from strongly regular graphs is the Krein condition. A strongly regular graph with parameters $(r^2(r+3)^2,\, r(r^2+3r+1),\, 0,\, r(r+1))$, the triangle-free parameter family attaining the Krein bound, would give $(d+1)^{4/3} + (2+o(1))(d+1)$ points, so the leading term is best possible if infinitely many such graphs exist, and $f(d) = O(d)$ would imply that only finitely many exist (the members $r = 3, 4$ are known not to exist).

We also prove $\operatorname{rank}(A - \lambda_2 I) > n/(8(\lambda_2+1))$ for every triangle-free graph with second largest eigenvalue $\lambda_2 > 0$, a lower bound for the rank in Problem 19 of Polyanskii, and $f(d) = O(d)$ for sets all of whose non-unit distances are at least some fixed $a > 1$. In the other direction, every connected regular triangle-free graph whose second largest eigenvalue is positive and has multiplicity m yields a two-distance almost-equidistant set of n points in $\mathbb{R}^{n-1-m}$; applied to the known triangle-free strongly regular graphs this gives $f(22) \geq 100$ and $f(d) > 2d+4$ for $20 \leq d \leq 57$, where $2d+4$ was the previous record. All constructions are certified by an exact, dependency-free verification script.

1. Introduction

For a positive integer d, a finite set $V \subset \mathbb{R}^d$ is *almost-equidistant* if among any three points of V some two are at Euclidean distance 1. Let $f(d)$ denote the largest cardinality of an almost-equidistant set in $\mathbb{R}^d$. The problem was raised by Z. Lángi and is a Euclidean analogue of Erdős's question on almost-orthogonal vectors, solved by Rosenfeld [25]. The vertex set of the Moser spindle shows $f(2) \geq 7$; Bezdek, Naszódi and Visy [5] proved $f(2) = 7$ and Györey proved $f(3) = 10$ (see [3]).

Balko, Pór, Scheucher, Swanepoel and Valtr [3] made a systematic study of f. They proved $f(d) \leq O(d^{3/2})$, later improved to $O(d^{13/9})$ by Polyanskii [23] and to $O(d^{4/3})$ by Kupavskii, Mustafa and Swanepoel [18] (see also [24]). On the lower bound side, [3] gives $f(d) \geq 2d+4$ for all $d \geq 3$ by an explicit construction, and, using the 16-point set of Larman and Rogers [20] in $\mathbb{R}^5$ derived from the Clebsch graph, $f(5) \geq 16$, $f(6) \geq 18$, $f(7) \geq 20$ and $f(9) \geq f(8) \geq 24$. The conjecture $f(d) = O(d)$ is stated in [23, Conjecture 12], and Polyanskii [24, Problems 18 and 19] asked for the largest almost-equidistant set that is at the same time a two-distance set, reducing this to a question about triangle-free graphs whose second largest eigenvalue has large multiplicity.

Everything in this paper goes through one object: the graph G on V whose edges are the pairs at non-unit distance. By definition G is triangle-free, its independent sets are unit simplices, and

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when V is a two-distance set with distances 1 and $a > 1$ the matrix $(\|v_i - v_j\|^2 - 1)_{ij}$ is $(a^2 - 1)$ times the adjacency matrix of G.

The main theorem. Call an almost-equidistant set *short-free* if no two of its points are at distance less than 1. Two-distance almost-equidistant sets with second distance above 1 are short-free, and so are the sets of Theorem 1.6 below; the direct sums and the extended sets of Section 7 in general are not.

1.1. **Theorem.** *Let $V \subset \mathbb{R}^d$ be a short-free almost-equidistant set and $D = d + 1$. Then*

$$\textit{(i)} \quad |V| \le D^{4/3} + 6D, \qquad\qquad \textit{(ii)} \quad |V| \le D^{4/3} + 2D + 100D^{2/3}.$$

By [24, Theorem 9] a two-distance almost-equidistant set whose second distance is below 1 has at most $2d + 4 = 2D + 2$ points, which is below both bounds, so both bounds hold for every two-distance almost-equidistant set.

The proof (Section 3) combines four exact relations for a short-free set with centroid at the origin: the row sums of the matrix $(\|v_i - v_j\|^2 - 1)$ are determined by the norms $\|v_i\|^2$; the far neighbourhood of each point is a unit simplex, which bounds the second moment of each row by its degree; the matrix $(\langle v_i, v_j\rangle^3)$ is a Gram matrix, so $\sum_{i,j}\langle v_i, v_j\rangle^3 \ge 0$; and the matrix $(\|v_i - v_j\|^2 - 1)$ has the eigenvalue 1 with multiplicity at least $n - d - 2$ (Polyanskii [23]). Hölder's inequality links the second and third moments of the rows, and the four relations then close into an inequality in n and d alone. Remark 3.4 lists the three places where the hypothesis that there are no short distances is used and explains why the argument does not extend beyond it.

The leading term of Theorem 1.1 cannot be improved unless a well-known family of strongly regular graphs is finite.

1.2. **Theorem.** *For an integer $r \ge 1$ let* Kre(r) *denote a strongly regular graph with parameters $(r^2(r+3)^2,\, r(r^2+3r+1),\, 0,\, r(r+1))$. These are exactly the parameter sets of triangle-free strongly regular graphs attaining the Krein bound with equality. If* Kre(r) *exists, it yields a two-distance almost-equidistant set of $n_r = r^2(r+3)^2$ points in $\mathbb{R}^d$ with $d = r(r^2+3r+1)$, and with $D = d+1$,*

$$n_r = D^{4/3} + 2D - \tfrac{1}{3}D^{2/3} + O(D^{1/3}).$$

Consequently, if Kre(r) *exists for infinitely many r, then along those dimensions the largest two-distance almost-equidistant set has between $D^{4/3}+2D-\frac{1}{3}D^{2/3}+O(D^{1/3})$ and $D^{4/3}+2D+100D^{2/3}$ points, so the first two terms of Theorem 1.1(ii) are best possible; and if $f(d) = O(d)$ then* Kre(r) *exists for only finitely many r.*

Kre(1) and Kre(2) are the Clebsch and Higman–Sims graphs; Kre(3) and Kre(4) do not exist, by Gavrilyuk and Makhnev [12] and Makhnev [22]; the existence of Kre(r) for $r \ge 5$ is open [10]. Section 4 proves Theorem 1.2 and explains why the hypothetical sets of Kre(r) sit exactly where the arguments of [18, 24] stop.

Two more upper bounds. The next result is a lower bound for the rank in Polyanskii's Problem 19.

1.3. **Theorem.** *Let G be a triangle-free graph on n vertices with adjacency matrix A, let $\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_n$ be its eigenvalues, and suppose $\lambda_2 > 0$. Put $r = \operatorname{rank}(A - \lambda_2 I)$. If $\lambda_1 > \lambda_2$ then $n < 8(\lambda_2 + 1)(r - 1)$, and if $\lambda_1 = \lambda_2$ then $n \le 2r$. In particular*

$$\operatorname{rank}(A - \lambda_2 I) > \frac{n}{8(\lambda_2 + 1)}.$$

1.4. **Corollary.** *Let $V \subset \mathbb{R}^d$ be a two-distance almost-equidistant set whose distances are 1 and a.*

(1) *If $a < 1$, then $|V| \le 2d + 4$ (Polyanskii* [24, Theorem 9]*).*
(2) *If $a > 1$, then $|V| < 8(\theta + 1)(d + 1)$ with $\theta = 1/(a^2 - 1)$, that is, $|V| < 8a^2(d + 1)/(a^2 - 1)$.*

(3) *If $a > 1$ and $|V| > 2d + 3$, then $a^2 = k/(k-1)$ for an integer k with $2 \le k \le \frac{1}{2} + \sqrt{d/2}$, the number $\theta = k - 1$ is a positive integer, and $|V| < 8k(d+1)$.*

Corollary 1.4 says that a sequence of two-distance almost-equidistant sets with $|V_d|/d \to \infty$ must have second distance tending to 1. Without the two-distance assumption we can only say the following.

1.5. **Theorem.** *There is an absolute constant C with the following property. Let $a > 1$ and let $V \subset \mathbb{R}^d$ be an almost-equidistant set in which every two points at non-unit distance are at distance at least a. Then*

$$|V| \le C\, d\, \max\{1, (a^2-1)^{-2}\}.$$

Theorem 1.5 confirms Conjecture 12 of [23] for sets all of whose non-unit distances are at least some fixed $a > 1$; it says nothing about sets with pairs at distance below 1, and a sequence with $|V_d|/d \to \infty$ must therefore contain non-unit pairs at distance less than $1 + \varepsilon$ for every $\varepsilon > 0$, above or below 1. Neither Theorem 1.3 nor Theorem 1.5 says anything in the regime of the sets of Theorem 1.2, whose second distance $\sqrt{1 + 1/r}$ tends to 1; Theorem 1.1 does.

Constructions. The use of strongly regular graphs to build two-distance sets with prescribed properties goes back at least to Bondarenko's counterexample to Borsuk's conjecture [6] and its refinement by Jenrich and Brouwer [17]. For almost-equidistant sets the first instance is the Larman–Rogers set from the Clebsch graph, and it is the first member of a family: the Clebsch graph is one of the seven known triangle-free strongly regular graphs (the pentagon, the Petersen, Clebsch, Hoffman–Singleton [16], Gewirtz [13], M_{22} [7] and Higman–Sims [15] graphs), and the argument of Larman and Rogers applies to each of them, and to every connected regular triangle-free graph whose second largest eigenvalue is positive.

1.6. **Theorem.** *Let H be a connected k-regular triangle-free graph on n vertices whose second largest adjacency eigenvalue λ_2 is positive and has multiplicity m. Then there is an almost-equidistant set of n points in $\mathbb{R}^{n-1-m}$, all lying on a sphere of radius r with*

$$r^2 = \frac{1}{2} + \frac{k - \lambda_2}{2n\lambda_2}.$$

Only two distances occur in this set, namely 1 and $\sqrt{1 + 1/\lambda_2}$.

Combined with an orthogonal direct-sum construction (Lemma 7.1) and an extension lemma (Lemma 7.5), Theorem 1.6 gives the following.

1.7. **Theorem.** *The following lower bounds hold:*

d	10	20	21	22	23	24	25	26	27	28	29	30	31
$f(d) \ge$	26	58	78	100	102	104	106	108	110	112	114	116	120
previous	24	44	46	48	50	52	54	56	58	60	62	64	66

Consequently, since f is non-decreasing, $f(d) \ge 120 > 2d + 4$ for $31 \le d \le 57$.

The bound $f(22) \ge 100$ comes from the Higman–Sims graph alone; $f(31) \ge 120$ adds to it the 20 vertices of a regular unit 9-simplex and its antipodal image, placed in an orthogonal $\mathbb{R}^9$. The two-distance set of 100 points in $\mathbb{R}^{22}$ shows that the minimum of $\operatorname{rank}(A - \lambda_2 I)$ over triangle-free graphs on 100 vertices with $\lambda_2 > 0$, the quantity in [24, Problem 19], is at most 23. These constructions are exhausted by the known triangle-free strongly regular graphs and finitely many direct sums, give nothing beyond $d = 57$, and say nothing about the growth of f; their ingredients are classical, and the contribution is the bookkeeping that turns them into verified numbers. Everything in Theorem 1.7 is certified by a standard-library Python script that rebuilds the graphs from the binary Golay code, checks all identities in integer and rational arithmetic, and tests every triple of points; the script is reproduced in Appendix A.

Organisation. Section 2 collects the matrix of excesses and the four relations behind Theorem 1.1, which is proved in Section 3. Section 4 treats the Krein family. Sections 5 and 6 prove Theorems 1.3 and 1.5. Section 7 contains the constructions and the proof of Theorem 1.7, and Section 8 collects remarks, the slack of each inequality of Section 3 on the known configurations, and open questions.

2. The matrix of excesses

Throughout, $\langle\cdot,\cdot\rangle$ is the standard inner product, J is the all-ones matrix and $\mathbf{1}$ the all-ones vector. We first record the matrix that Polyanskii attached to an almost-equidistant set; the lemma is [24, Lemma 1 and Corollary 5] and we include the short proof.

2.1. **Lemma** (Polyanskii)**.** *Let $V=\{v_1,\dots,v_n\}$ be an almost-equidistant set in $\mathbb{R}^d$ and let U be the $n\times n$ matrix with entries $U_{ij}=\|v_i-v_j\|^2-1$ for $i\neq j$ and $U_{ii}=0$. Then U has at least $n-d-2$ eigenvalues equal to 1, at most one eigenvalue larger than 1, and $\operatorname{tr}U=\operatorname{tr}U^3=0$. If no eigenvalue of U exceeds 1, then $n\le 2d+4$.*

Proof. Let $\Gamma=(\langle v_i,v_j\rangle)_{ij}$, which is positive semidefinite of rank at most d, and let c be the vector with $c_i=\|v_i\|^2$. Then $\|v_i-v_j\|^2=c_i+c_j-2\langle v_i,v_j\rangle$, so with $x=c-\frac12\mathbf{1}$,

$$U-I=-2\Gamma+x\mathbf{1}^\top+\mathbf{1}x^\top.$$

The right-hand side has rank at most $d+2$, so 1 is an eigenvalue of U of multiplicity at least $n-d-2$. The matrix $x\mathbf{1}^\top+\mathbf{1}x^\top$ has at most one positive eigenvalue (its non-zero eigenvalues are $\langle x,\mathbf{1}\rangle\pm\|x\|\|\mathbf{1}\|$, whose product is at most 0), and -2Γ is negative semidefinite, so by Weyl's inequality $U-I$ has at most one positive eigenvalue. The diagonal of U vanishes, so $\operatorname{tr}U=0$, and $(U^3)_{ii}=\sum_{j,l}U_{ij}U_{jl}U_{li}$ vanishes because a non-zero term would be a triple of points pairwise at non-unit distance. The last statement is [24, Corollary 5]: if the eigenvalues of U are 1 with multiplicity $m\ge n-d-2$ and $\lambda_1,\dots,\lambda_{n-m}\le 1$, then $\operatorname{tr}U=0$ and $\operatorname{tr}U^3=0$ give $\sum(-\lambda_i)=m=\sum(-\lambda_i)^3$ with $-\lambda_i\ge -1$, and [24, Lemma 3] shows that this forces $n-m\ge m$, i.e. $n\le 2(n-m)\le 2d+4$. □

2.2. **Proposition.** *Let $V=\{v_1,\dots,v_n\}\subset\mathbb{R}^d$ be a two-distance almost-equidistant set with distances 1 and $a>1$, let G be the graph of the pairs at distance a, and let $\theta=1/(a^2-1)$. Then G is triangle-free, θ is an eigenvalue of A of multiplicity at least $n-d-2$, and at most one eigenvalue of A exceeds θ. In particular, if $n\ge d+4$ then $\theta=\lambda_2$ and $\operatorname{rank}(A-\lambda_2 I)\le d+2$.*

Conversely, let G be a triangle-free graph on $[n]$, let $\theta>0$ and $P=I-\frac1n J$. If $P(\theta I-A)P$ is positive semidefinite, then the n points with Gram matrix $\frac{1}{2\theta}P(\theta I-A)P$ form a two-distance almost-equidistant set in $\mathbb{R}^d$, $d=\operatorname{rank}P(\theta I-A)P$, whose pairs at distance $\sqrt{1+1/\theta}$ are the edges of G and whose other pairs are at distance 1.

Proof. For the first part, $U=(a^2-1)A$ in the notation of Lemma 2.1, so $A=\theta U$ has the eigenvalue θ with multiplicity at least $n-d-2$ and at most one eigenvalue above θ. If $n\ge d+4$ the multiplicity is at least 2, so θ is the second largest eigenvalue whether or not an eigenvalue exceeds it, and $\operatorname{rank}(A-\theta I)=n-\operatorname{mult}(\theta)\le d+2$.

For the converse, let $\Gamma=\frac{1}{2\theta}P(\theta I-A)P\succeq 0$ and let $p_1,\dots,p_n\in\mathbb{R}^d$ have Gram matrix Γ. A direct computation using $(PAP)_{ii}+(PAP)_{jj}-2(PAP)_{ij}=-2A_{ij}$ and $P_{ii}+P_{jj}-2P_{ij}=2$ gives $\|p_i-p_j\|^2=\Gamma_{ii}+\Gamma_{jj}-2\Gamma_{ij}=1+A_{ij}/\theta$ for $i\neq j$. Three points with pairwise distance $\sqrt{1+1/\theta}$ would be a triangle of G. □

The first part of the proposition is the reduction of Problem 18 to Problem 19 in [24]. For a regular graph $\mathbf{1}$ is an eigenvector of A, so $P(\theta I-A)P$ is $\theta I-A$ on $\mathbf{1}^\perp$ and the converse reduces to Theorem 1.6.

Notation for short-free sets. For the rest of this section and for Section 3, $V = \{v_1, \dots, v_n\} \subset \mathbb{R}^d$ is a short-free almost-equidistant set with $n \geq 2$, $D = d+1$, and $\sum_i v_i = 0$ (translate if necessary). We write

$$\|v_i\|^2 = \tfrac{1}{2} + a_i, \qquad X = \sum_i a_i, \qquad A = \max_i |a_i|.$$

G is the graph of the pairs at non-unit distance, $N(i)$ the neighbourhood of i in G and $k_i = |N(i)|$. For $i \neq j$ put $\eta_{ij} = \|v_i - v_j\|^2 - 1$, so that $\eta_{ij} \geq 0$ with equality exactly on the unit pairs; this is the matrix U of Lemma 2.1. The row sums of the first three powers are

$$F_i = \sum_{j \in N(i)} \eta_{ij}, \qquad s_i = \sum_{j \in N(i)} \eta_{ij}^2, \qquad t_i = \sum_{j \in N(i)} \eta_{ij}^3,$$

with totals $E_1 = \sum_i F_i$, $E_2 = \sum_i s_i = \|U\|_F^2$, $E_3 = \sum_i t_i$, and we put $T = \sum_i a_i s_i$ and $F = \max\{1, \max_i F_i\}$. Since G is triangle-free, $N(i)$ is a set of points at pairwise unit distance, hence affinely independent, so $k_i \leq D$; for the same reason the independence number of G is at most D.

2.3. **Lemma** (Row sums). (1) *$F_i = na_i + X + 1$ for every i, and $E_1 = n(2X+1)$; in particular $X \geq -\frac{1}{2}$.*
(2) *$X \leq \frac{1}{2}(F-1)$ and $|a_i| \leq F/n$ for every i.*
(3) *$F_i^2 \leq k_i s_i \leq D s_i$, $s_i^2 \leq F_i t_i$, $F_i s_i \leq k_i t_i$ and $E_2^2 \leq E_1 E_3$.*

Proof. (1) Since $\sum_j v_j = 0$, $\sum_j \|v_i - v_j\|^2 = n\|v_i\|^2 + \sum_j \|v_j\|^2 = n(\frac{1}{2} + a_i) + \frac{n}{2} + X$, and subtracting $n-1$ (the unit pairs contribute 0 to F_i) gives $F_i = na_i + X + 1$. Summing over i gives $E_1 = 2nX + n$, and $E_1 \geq 0$.

(2) $X = (E_1 - n)/(2n) \leq (nF - n)/(2n)$. From (1), $a_i = (F_i - X - 1)/n \leq (F - \frac{1}{2})/n < F/n$, and $-a_i = (X + 1 - F_i)/n \leq (X+1)/n \leq (F+1)/(2n) \leq F/n$ as $F \geq 1$.

(3) All four are the Cauchy–Schwarz inequality for non-negative numbers: $F_i = \sum \eta_{ij} \leq \sqrt{k_i}\sqrt{s_i}$; $s_i = \sum \eta_{ij}^{1/2} \eta_{ij}^{3/2} \leq \sqrt{F_i t_i}$, and likewise $E_2 \leq \sqrt{E_1 E_3}$; finally $t_i \geq s_i^2/F_i \geq s_i^2/\sqrt{k_i s_i} = s_i^{3/2}/\sqrt{k_i}$ and $F_i s_i \leq \sqrt{k_i}\, s_i^{3/2} \leq k_i t_i$. □

2.4. **Lemma** (The local bound). *For every i,*

$$s_i \leq (1 + 2A)(1 + 2Ak_i) + 4AF_i.$$

Proof. The points v_j, $j \in N(i)$, are at pairwise unit distance, so their Gram matrix is $\Gamma_i = \frac{1}{2}I + \frac{1}{2}(x\mathbf{1}^\top + \mathbf{1}x^\top)$ with $x = (a_j)_{j \in N(i)}$, and $\lambda_{\max}(\Gamma_i) \leq \frac{1}{2} + Ak_i$ because the eigenvalues of $x\mathbf{1}^\top + \mathbf{1}x^\top$ are at most $2\|x\|\sqrt{k_i} \leq 2Ak_i$. Let M be the matrix whose columns are these v_j and $c = M^\top v_i = (\langle v_i, v_j \rangle)_{j \in N(i)}$, so that $c_j = \frac{1}{2}(a_i + a_j - \eta_{ij})$. Then $\|c\|^2 \leq \lambda_{\max}(M^\top M)\|v_i\|^2 = \lambda_{\max}(\Gamma_i)\|v_i\|^2 \leq (\frac{1}{2} + Ak_i)(\frac{1}{2} + A)$. Since $4c_j^2 \geq \eta_{ij}^2 - 2(a_i + a_j)\eta_{ij} \geq \eta_{ij}^2 - 4A\eta_{ij}$, summing over j gives $s_i \leq 4\|c\|^2 + 4AF_i$. □

The third relation is the one that is new here.

2.5. **Lemma** (The third moment). *Put $L = 12\sum_i a_i^2 + 2n\sum_i a_i^3 + 6X\sum_i a_i^2$. Then*

$$E_3 \leq n + 6X + L + 6T, \qquad 6X \leq 3(F-1), \qquad L \leq \frac{1.153\,(F+1)^3 + \frac{4}{3}(F+1)^2}{n}.$$

Proof. Write $y_{ij} = \frac{1}{2}(a_i + a_j)$. Then $\langle v_i, v_i\rangle = \frac{1}{2} + a_i$ and, for $i \neq j$, $\langle v_i, v_j\rangle = \frac{1}{2}(\|v_i\|^2 + \|v_j\|^2 - \|v_i - v_j\|^2) = y_{ij} - \frac{1}{2}\eta_{ij}$, where $\eta_{ij} = 0$ on unit pairs. Expanding the cubes,

$$8\sum_i \langle v_i, v_i\rangle^3 = n + 6X + 12\sum a_i^2 + 8\sum a_i^3,$$

$$8\sum_{i\neq j} y_{ij}^3 = \sum_{i\neq j}(a_i + a_j)^3 = (2n-8)\sum a_i^3 + 6X\sum a_i^2,$$

$$8\sum_{i\neq j}\big(y_{ij} - \tfrac{1}{2}\eta_{ij}\big)^3 = 8\sum_{i\neq j} y_{ij}^3 - 12\sum_{i\neq j} y_{ij}^2\eta_{ij} + 6\sum_{i\neq j} y_{ij}\eta_{ij}^2 - E_3,$$

and $\sum_{i\neq j} y_{ij}\eta_{ij}^2 = \sum_i a_i s_i = T$ by symmetry. Adding up,

$$8\sum_{i,j}\langle v_i, v_j\rangle^3 = n + 6X + L + 6T - E_3 - 12\sum_{i\neq j} y_{ij}^2\eta_{ij}. \tag{1}$$

The matrix $(\langle v_i, v_j\rangle^3)_{ij}$ is the Gram matrix of the vectors $v_i \otimes v_i \otimes v_i$, so the left-hand side is non-negative, and the last sum is non-negative because $\eta_{ij} \geq 0$. This proves the first inequality.

For the other two put $c = X + 1$ and $g_i = F_i$, so that $a_i = (g_i - c)/n$ by Lemma 2.3(1), $\frac{1}{2} \leq c \leq \frac{1}{2}(F+1) \leq F$ by Lemma 2.3(2), and $6X = 6c-6 \leq 3(F-1)$. Since $(g_i-c)^3 \leq (F-c)(g_i-c)^2$ for every i,

$$L = 6(c+1)\sum a_i^2 + 2n\sum a_i^3 \leq \frac{6(c+1) + 2(F-c)}{n^2}\sum_i (g_i - c)^2 = \frac{2F + 4c + 6}{n^2}\sum_i (g_i - c)^2.$$

Here $\sum_i (g_i - c)^2 = \sum g_i^2 - 2cE_1 + nc^2 \leq FE_1 - 2cE_1 + nc^2 = n(2Fc - F - 3c^2 + 2c)$, using $g_i \leq F$ and $E_1 = n(2c-1)$. Writing $c = \gamma(F+1)$ with $0 < \gamma \leq \frac{1}{2}$, we have $2Fc - F - 3c^2 + 2c \leq 2c(F+1) - 3c^2 = (F+1)^2\gamma(2-3\gamma)$ and $2F + 4c + 6 = (2+4\gamma)(F+1) + 4$, so

$$L \leq \frac{(F+1)^3\,\gamma(2+4\gamma)(2-3\gamma) + 4(F+1)^2\,\gamma(2-3\gamma)}{n}.$$

The function $\gamma(2+4\gamma)(2-3\gamma) = 4\gamma + 2\gamma^2 - 12\gamma^3$ has derivative $4 + 4\gamma - 36\gamma^2$, which vanishes at $\gamma^* = (1 + \sqrt{37})/18$, and its maximum $4\gamma^* + 2\gamma^{*2} - 12\gamma^{*3} < 1.153$; and $\gamma(2-3\gamma) \leq \frac{1}{3}$. □

For the sets of Theorem 1.6 built from a strongly regular graph the inequality $\sum_{i,j}\langle v_i, v_j\rangle^3 \geq 0$ is the Krein condition of the graph (Remark 7.4); this is why the Krein-tight family of Section 4 is where Lemma 2.5 is tight.

2.6. **Lemma** (The rank bound)**.** *Let $m = n - D - 1$ and $\mu = m/(D+1)$.*

1. *If $n \geq D + 1$ then $E_2 \geq m + \frac{m^2}{D+1} = n\mu$.*
2. *If $n > 2D + 2$, let λ_0 be the largest eigenvalue of U. Then $\lambda_0 > 1$, $\lambda_0^3 \geq (n-D)^3/D^2 - n$, and*

$$E_2 \geq m + \lambda_0^2 + \frac{(m + \lambda_0)^2}{D}.$$

Proof. By Lemma 2.1, U has the eigenvalue 1 with multiplicity $m' \geq m$, trace 0, and $E_2 = \|U\|_F^2$ is the sum of the squares of its eigenvalues.

(1) The remaining $n - m' \leq D + 1$ eigenvalues sum to $-m'$, so by the Cauchy–Schwarz inequality their squares sum to at least $m'^2/(n-m') \geq m'^2/(D+1)$. Hence $E_2 \geq m' + m'^2/(D+1) \geq m + m^2/(D+1)$, and $m + m^2/(D+1) = m(m+D+1)/(D+1) = mn/(D+1) = n\mu$.

(2) If $n > 2D + 2$ then some eigenvalue of U exceeds 1 by Lemma 2.1, and it is unique; call it λ_0. The other $n - 1$ eigenvalues are at most 1 and at least m of them equal 1; label m of them as 1 and the remaining D as $\lambda_1, \ldots, \lambda_D$. Then $\operatorname{tr} U = \operatorname{tr} U^3 = 0$ read $\sum_{i=1}^D(-\lambda_i) = \lambda_0 + m$ and $\sum_{i=1}^D(-\lambda_i)^3 = \lambda_0^3 + m$, with $-\lambda_i \geq -1$ and $\lambda_0 + m > D$, so [24, Lemma 3] gives $\sum(-\lambda_i)^3 \geq$

$(\lambda_0+m)^3/D^2 \ge (n-D)^3/D^2$ and therefore $\lambda_0^3 \ge (n-D)^3/D^2 - m \ge (n-D)^3/D^2 - n$; this is [24, Lemma 2(3)]. For the Frobenius norm, the eigenvalues other than λ_0 and other than the m' ones are at most D in number and sum to $-m'-\lambda_0$, so $E_2 \ge \lambda_0^2 + m' + (m'+\lambda_0)^2/D \ge \lambda_0^2 + m + (m+\lambda_0)^2/D$. □

3. Proof of Theorem 1.1

We keep the notation of Section 2. The lemmas there close into one inequality in which only n, D, E_2 and the quantity

$$R := 3(F-1) + \frac{1.153\,(F+1)^3 + \frac{4}{3}(F+1)^2}{n}$$

appear; by Lemma 2.5, $n + 6X + L \le n + R$.

3.1. **Proposition.** *Let V be a short-free almost-equidistant set of n points in $\mathbb{R}^d$ with $n > 6D$, and put $u = D/n$. Then for every $0 \le E \le E_2$,*

$$(1-6u)\Big[\Big(\frac{E}{n}\Big)^{3/2} D^{-1/2} + \frac{3(E/n)^3}{n+R}\Big] \le 1 + \frac{R}{n}. \tag{2}$$

Proof. By Lemma 2.3(1), $T = \frac{1}{n}\sum_i (F_i - X - 1)s_i$, so

$$E_3 - 6T = \sum_i \Big[t_i - \frac{6}{n}F_i s_i\Big] + \frac{6(X+1)}{n}E_2. \tag{3}$$

Fix i with $s_i > 0$. By Lemma 2.3(3), $t_i \ge s_i^2/F_i$ and $0 < F_i \le \sqrt{Ds_i}$, and the function $\phi \mapsto s_i^2/\phi - \frac{6}{n}\phi s_i$ is decreasing on $\phi > 0$; hence

$$t_i - \frac{6}{n}F_i s_i \ge \frac{s_i^2}{\sqrt{Ds_i}} - \frac{6}{n}\sqrt{Ds_i}\,s_i = (1-6u)\,\frac{s_i^{3/2}}{\sqrt{D}},$$

and this also holds when $s_i = 0$. Next, $X + 1 = (E_1 + n)/(2n)$ and $E_1 \ge E_2^2/E_3$ by Lemma 2.3, so

$$\frac{6(X+1)}{n}E_2 = \frac{3(E_1+n)E_2}{n^2} \ge \frac{3E_2^3}{n^2 E_3}.$$

On the other hand $E_3 \le n + R + 6T$ by Lemma 2.5, and $T \le \frac{1}{n}\sum_i F_i s_i \le \frac{1}{n}\sum_i k_i t_i \le uE_3$ by Lemma 2.3(3) and $k_i \le D$ (we used $X + 1 > 0$); hence $E_3 \le (n+R)/(1-6u)$, and so

$$\frac{6(X+1)}{n}E_2 \ge \frac{3(1-6u)E_2^3}{n^2(n+R)}.$$

Inserting both estimates into (3) and using $E_3 - 6T \le n + R$ once more,

$$(1-6u)\frac{\sum_i s_i^{3/2}}{\sqrt{D}} + \frac{3(1-6u)E_2^3}{n^2(n+R)} \le n + R.$$

By the power mean inequality $\sum_i s_i^{3/2} \ge n(E_2/n)^{3/2}$. Dividing by n gives (2) with $E = E_2$, and the left-hand side of (2) is increasing in E. □

3.2. **Lemma** (A priori bound on F)**.** *In the setting of Proposition 3.1, $F^2(1-4u-4u^2) \le D + 2(D+1)uF$. Consequently, if $u \le u_0$ and $0 < q_0 \le 1 - 4u_0 - 4u_0^2$, then*

$$F \le \frac{2(D+1)u_0}{q_0} + \sqrt{\frac{D}{q_0}}.$$

Proof. If $F = 1$ the first inequality holds because its right-hand side is at least $D \ge 1$. Otherwise let i attain $F = F_i$. By Lemma 2.3(3), Lemma 2.4, $A \le F/n$ (Lemma 2.3(2)) and $k_i \le D$,

$$F^2 \le Ds_i \le D\Big[\Big(1+\frac{2F}{n}\Big)\Big(1+\frac{2FD}{n}\Big)+\frac{4F^2}{n}\Big] = D + 2(D+1)uF + 4u^2F^2 + 4uF^2.$$

For the second statement, $q_0F^2 - 2(D+1)u_0F - D \le 0$, so F is at most the larger root, $F \le \big[(D+1)u_0 + \sqrt{(D+1)^2u_0^2 + Dq_0}\big]/q_0 \le 2(D+1)u_0/q_0 + \sqrt{D/q_0}$. □

Proof of Theorem 1.1(i). If $D \le 13$, the far graph G is triangle-free with independence number at most D, so $n < R(3, D+1) \le \binom{D+2}{2}$ by the Erdős–Szekeres bound, and $\binom{D+2}{2} - 1 \le D^{4/3} + 6D$ for $1 \le D \le 13$ (for $D = 13$ the two sides are 104 and 108.5). So let $D \ge 14$ and put $t = D^{1/3} > 2.41$.

A subset of a short-free almost-equidistant set is again one, so it suffices to show that no such set has exactly $n = N$ points, where N is the least integer with $N \ge n_0 := D^{4/3} + 6D = t^3(t+6)$. Suppose it does. Then $u = D/n \le u_0 := 1/(t+6) < 0.119$, so $4u^2 \le 0.476u$ and $1 - 4u - 4u^2 \ge 1 - 4.48u_0 = (t+1.52)/(t+6) =: q_0$, and Lemma 3.2 gives

$$F \le \frac{2(t^3+1)}{t+1.52} + t^{3/2}\sqrt{\frac{t+6}{t+1.52}}. \tag{4}$$

We claim that $F + 1 \le 2.7t^2$. Dividing (4) by t^2 and using $t^{-2} \le 0.173$ and $\sqrt{(t+6)/(t+1.52)} \le 1.463$ for $t \ge 2.41$,

$$\frac{F+1}{t^2} \le \frac{2(t+0.173)}{t+1.52} + \frac{1.463}{\sqrt{t}} + \frac{1}{t^2}.$$

The first term is increasing in t and the other two are decreasing, so on $[2.41, 4]$, $[4, 9]$ and $[9, \infty)$ the right-hand side is at most $1.513 + 0.943 + 0.173$, $1.745 + 0.732 + 0.063$ and $2 + 0.488 + 0.013$ respectively, each below 2.7. Hence, with $n \ge t^3(t+6)$,

$$R \le 3(F+1) + \frac{1.153(F+1)^3 + \frac{4}{3}(F+1)^2}{n} \le 8.1t^2 + \frac{22.7t^6 + 9.72t^4}{t^3(t+6)} = \frac{tP(t)}{t+6},$$
$$P(t) := 30.8t^2 + 48.6t + 9.72.$$

Now apply Proposition 3.1 with $E = n\mu$, $\mu = (n-D-1)/(D+1)$, which is allowed by Lemma 2.6(1). Since $n \ge n_0$,

$$\mu \ge \mu_0 := \frac{t^4 + 5t^3 - 1}{t^3 + 1} \ge t + 4.3,$$

the last inequality being $0.7t^3 \ge t + 5.3$, true for $t \ge 2.41$. Also $1 - 6u \ge 1 - 6u_0 = t/(t+6)$, and $\mu^{3/2}D^{-1/2} = (\mu/t)^{3/2} \ge (1 + 4.3/t)^{3/2} \ge 1 + 6.45/t$, so the first term in the bracket of (2) is at least $(t+6.45)/(t+6) > 1$. For the second term, $\psi(n) = (n-D-1)^3/(n+R)$ is increasing in n (its derivative has the sign of $3(n+R) - (n-D-1) > 0$), so

$$\frac{3\mu^3}{n+R} \ge \frac{3\mu_0^3}{n_0 + tP(t)/(t+6)} \ge \frac{3(t+4.3)^3(t+6)}{t^3(t+6)^2 + tP(t)},$$

while $R/n \le tP(t)/((t+6)n_0) = P(t)/(t^2(t+6)^2)$. Inequality (2) therefore implies

$$\frac{t}{t+6} \cdot \frac{3(t+4.3)^3(t+6)}{t^3(t+6)^2 + tP(t)} < \frac{P(t)}{t^2(t+6)^2}, \qquad \text{i.e.} \qquad 3t^2(t+4.3)^3(t+6)^2 < P(t)\big(t^2(t+6)^2 + P(t)\big).$$

But the polynomial $p(t) = 3t^2(t+4.3)^3(t+6)^2 - P(t)(t^2(t+6)^2 + P(t))$ has degree 7 with leading coefficient 3, and expanding $p(s+2.41)$ in powers of s one finds that all eight coefficients are positive (the constant term is $p(2.41) > 1.5 \cdot 10^5$ and the left-hand side above exceeds the right-hand side by a factor larger than 1.69 at $t = 2.41$). So $p(t) > 0$ for all $t \ge 2.41$, a contradiction. □

3.3. **Remark.** The constant 6 comes from rounding at every step. Evaluating the chain of Proposition 3.1 and Lemmas 3.2 and 2.6(1) numerically, without the roundings, excludes $n > D^{4/3}+5.75D$ for every $2 \le D \le 300$ and $n > D^{4/3}+3.02D$ for $D = 10^9$; with Lemma 2.6(2) in place of (1) the corresponding values are 4.75 and 2.02. The ancillary file `chain_bounds.py` tabulates these values.

Proof of Theorem 1.1(ii). For $D \le 15625$ we have $4D \le 100D^{2/3}$, so (ii) follows from (i). Let $D \ge 8000$, $t = D^{1/3} \ge 20$, and suppose, as before, that a short-free almost-equidistant set has exactly n points, where $n_1 \le n < n_1 + 1$ and $n_1 := D^{4/3} + 2D + 100D^{2/3} = t^4 + 2t^3 + 100t^2$. We use Proposition 3.1 with the lower bound of Lemma 2.6(2). Throughout, $t \ge 20$ is used to absorb lower-order terms.

The quantities u, F, R. We have $u = t^3/n \le 1/(t + 2 + 100/t) \le 1/(t+2) \le 1/22$. Writing $6u = \frac{6}{t} \cdot \frac{1}{1+y}$ with $y = 2/t + 100/t^2 \le 0.35$ and using $1/(1+y) \le 1 - y + y^2$ and $y^2 \le 49/t^2$,

$$1 - 6u \ge 1 - \frac{6}{t} + \frac{12}{t^2} + \frac{600 - 294}{t^3} \ge 1 - \frac{6}{t} + \frac{12}{t^2}. \tag{5}$$

Since $4u^2 \le 0.18u$, Lemma 3.2 applies with $u_0 = 1/(t+2)$ and $q_0 = 1 - 4.2u_0 = (t-2.2)/(t+2)$:

$$F \le \frac{2(t^3+1)}{t-2.2} + t^{3/2}\sqrt{\frac{t+2}{t-2.2}} \le \frac{2t^2(1+t^{-3})}{1-2.2/t} + 1.112\,t^{3/2} \le 2.248t^2 + 0.249t^2,$$

so $F + 1 \le 2.51t^2$, and with $n \ge t^4$,

$$R \le 3(F+1) + \frac{1.153(F+1)^3 + \frac{4}{3}(F+1)^2}{t^4} \le 7.53t^2 + 18.23t^2 + 8.4 \le 26t^2, \qquad \frac{R}{n} \le \frac{26}{t^2}. \tag{6}$$

The eigenvalue λ_0. By Lemma 2.6(2), with $n - D \ge n_1 - D = t^2(t^2 + t + 100)$ and $n < n_1 + 1$,

$$\lambda_0^3 \ge (t^2+t+100)^3 - (t^4 + 2t^3 + 100t^2 + 1) \ge (t^2+t)^3 + 300t^4 - (t^4 + 2t^3 + 100t^2 + 1) \ge (t^2+t)^3,$$

so $\lambda_0 \ge t^2 + t$.

The lower bound for E_2. Put $m = n - D - 1 \ge t^4 + t^3 + 100t^2 - 1$, so that $m/D \ge t + 1 + 100/t - t^{-3}$. By Lemma 2.6(2),

$$\mu' := \frac{E_2}{n} \;\ge\; \frac{m}{n} + \frac{\lambda_0^2}{n} + \frac{m^2}{nD} + \frac{2m\lambda_0}{nD} = \frac{m(n-1)}{nD} + \frac{\lambda_0^2}{n} + \frac{2m\lambda_0}{nD}.$$

We bound the three terms separately, using $n \le n_1 + 1 \le t^4(1 + 2/t + 100/t^2 + t^{-4})$ and $n \ge t^4$:

$$\begin{aligned}
\frac{m(n-1)}{nD} &\ge \frac{m}{D}\Big(1 - \frac{1}{t^4}\Big) \ge t\Big(1 + \frac{1}{t} + \frac{100}{t^2} - \frac{1}{t^4}\Big)\Big(1 - \frac{1}{t^4}\Big) \ge t\Big(1 + \frac{1}{t} + \frac{99.9}{t^2}\Big),\\
\frac{\lambda_0^2}{n} &\ge \frac{t^4(1+1/t)^2}{t^4(1+2/t+100/t^2+t^{-4})} = 1 - \frac{99 + t^{-2}}{t^2(1+2/t+100/t^2+t^{-4})} \ge 1 - \frac{100}{t^2},\\
\frac{2m\lambda_0}{nD} &\ge \frac{2(t^4+t^3+100t^2-1)(t^2+t)}{t^7(1+2/t+100/t^2+t^{-4})} \ge \frac{2}{t} \cdot \frac{1+2/t+100/t^2}{1+2/t+100/t^2+t^{-4}} \ge \frac{2}{t} - \frac{2}{t^5},
\end{aligned}$$

where in the first line we used $(1 + 1/t + 100/t^2)t^{-4} + t^{-4} \le 0.1t^{-2}$ and in the third $(t^4 + t^3 + 100t^2 - 1)(t^2 + t) \ge t^6(1 + 2/t + 100/t^2)$. Adding up,

$$\mu' \ge t + 2 + \frac{101.9}{t} - \frac{100}{t^2} - \frac{2}{t^5} \ge t\Big(1 + \frac{2}{t} + \frac{96}{t^2}\Big) =: t(1+x).$$

The two terms of (2). First, $(\mu'/t)^{3/2} \ge (1+x)^{3/2} \ge 1 + \frac{3}{2}x = 1 + 3/t + 144/t^2$. Second, $\mu'^3 \ge t^3(1+x)^3 \ge t^3(1+3x) = t^3(1 + 6/t + 288/t^2)$ and, by (6), $n + R \le t^4(1 + 2/t + 126/t^2 + t^{-4}) \le t^4(1 + 2/t + 126.01/t^2)$, so with $1/(1+y) \ge 1 - y$,

$$\frac{3\mu'^3}{n+R} \ge \frac{3}{t}\Big(1 + \frac{6}{t} + \frac{288}{t^2}\Big)\Big(1 - \frac{2}{t} - \frac{126.01}{t^2}\Big) \ge \frac{3}{t}\Big(1 + \frac{4}{t} + \frac{149.9}{t^2} - \frac{1333}{t^3} - \frac{36291}{t^4}\Big) \ge \frac{3}{t} + \frac{12}{t^2} - \frac{22.5}{t^3},$$

where the last step uses $t \geq 20$ twice. Hence the bracket in (2) is at least $1+6/t+156/t^2-22.5/t^3$, and with (5),

$$(1-6u)\Big[\dots\Big] \geq \Big(1-\frac{6}{t}+\frac{12}{t^2}\Big)\Big(1+\frac{6}{t}+\frac{156}{t^2}-\frac{22.5}{t^3}\Big) \geq 1+\frac{132}{t^2}-\frac{886.5}{t^3},$$

as one sees by multiplying out (the terms of order t^{-4} and t^{-5} add up to $2007t^{-4}-270t^{-5}>0$). By (6) the right-hand side of (2) is at most $1+26/t^2$, so (2) gives $106/t^2 < 886.5/t^3$, i.e. $t<8.4$, a contradiction. □

3.4. **Remark.** The hypothesis that V is short-free enters in three places: in Lemma 2.3(1), through $E_1 \geq 0$, which gives $X \geq -\frac{1}{2}$ and hence the positivity of $X+1$ used in Lemma 2.5 and in Proposition 3.1; in Lemma 2.5, where the term $-12\sum y_{ij}^2\eta_{ij}$ of (1) is dropped; and in Lemma 2.3(3), where Hölder's inequality is applied to the non-negative numbers η_{ij}. The second use is harmless, since for pairs at distance below 1 the dropped term is at most $12A^2\sum|\eta_{ij}|$, which is of lower order, and the first only restricts the range of X. The third is the obstacle: a pair at distance $a<1$ contributes $(1-a^2)^2$ to E_2 and $-(1-a^2)^3$ to E_3, so the third moment no longer controls the second, and since each far neighbourhood is a unit simplex such pairs may contribute up to $nD(1-a^2)^2$ to E_2, which exceeds the $nD^{1/3}$ that Lemma 2.6 demands as soon as $1-a^2 \gg D^{-1/3}$. So Theorem 1.1 says nothing about almost-equidistant sets with pairs at distance below 1, and extending it to them, even with $1-a^2$ bounded below by a constant, would require a way of controlling the short pairs by something other than the Frobenius norm of U. The identity $\operatorname{tr}U^3=0$ does control them, and is how [18, 24] obtain $O(d^{4/3})$ for all almost-equidistant sets, but it gives no information about the constant.

4. The Krein family

A strongly regular graph with parameters (v,k,λ,μ) has eigenvalues $k>\theta>\tau$; for the facts used below see [14, Chapter 10] or [8]. Corollary 7.3 in Section 7 shows that every triangle-free strongly regular graph yields a two-distance almost-equidistant set of v points in $\mathbb{R}^{m_\tau}$, where m_τ is the multiplicity of τ.

4.1. **Proposition.** *For an integer $r \geq 1$ put*

$$k_r = r(r^2+3r+1), \qquad \mu_r = r(r+1), \qquad n_r = r^2(r+3)^2.$$

(1) *Every strongly regular graph with parameters $(v,k,0,\mu)$ and eigenvalues $k>\theta>\tau$ satisfies $-\tau \leq \theta^2+2\theta$. A strongly regular graph with parameters $(n_r,k_r,0,\mu_r)$ has eigenvalues k_r, r and $-r(r+2)$ with multiplicities 1, n_r-1-k_r and k_r, and attains equality: $-\tau=\theta^2+2\theta$. Conversely every triangle-free strongly regular graph with equality has these parameters for some r (with r allowed to be a positive real number in the conference case).*

(2) *If a strongly regular graph with parameters $(n_r,k_r,0,\mu_r)$ exists, then $f(k_r) \geq n_r$, the n_r points lie on a sphere of squared radius $\frac{1}{2}+\frac{1}{2r(r+3)}$, and with $D=k_r+1$,*

$$n_r = D^{4/3}+2D-\tfrac{1}{3}D^{2/3}+O(D^{1/3}).$$

(3) *Consequently, if graphs with parameters $(n_r,k_r,0,\mu_r)$ exist for infinitely many r, then the bound $f(d)=O(d^{4/3})$ of* [18] *is sharp up to the constant, and Theorem 1.1(ii) is sharp in its first two terms; and Conjecture 12 of* [23]*, $f(d)=O(d)$, implies that such graphs exist for only finitely many r.*

Proof. (1) For a strongly regular graph with $\lambda=0$ the eigenvalue relations read $\theta+\tau=-\mu$ and $\theta\tau=\mu-k$, so $k=-(\theta+\tau)-\theta\tau$. The second Krein condition is $(\tau+1)(k+\tau+2\theta\tau) \leq (k+\tau)(\theta+1)^2$. Substituting, $k+\tau=-\theta(1+\tau)$ and $k+\tau+2\theta\tau=\theta(\tau-1)$, so the condition becomes $\theta(\tau^2-1) \leq -\theta(\tau+1)(\theta+1)^2$, i.e. $(\tau+1)\big(\tau-1+(\theta+1)^2\big) \leq 0$. Since $\tau+1<0$, this is $-\tau \leq (\theta+1)^2-1 = \theta^2+2\theta$. If equality holds, put $\theta=r$; then $\tau=-r(r+2)$, $\mu=-(\theta+\tau)=r(r+1)=\mu_r$,

$k = -(\theta+\tau) - \theta\tau = r(r+1) + r^2(r+2) = k_r$, and $v = 1 + k + k(k-1)/\mu = n_r$ by a direct calculation ($k_r(k_r-1) = r(r^2+3r+1)(r^3+3r^2+r-1)$, and $r^3+3r^2+r-1 = (r+1)(r^2+2r-1)$, so $k_r(k_r-1)/\mu_r = (r^2+3r+1)(r^2+2r-1)$, whence $1 + k_r + k_r(k_r-1)/\mu_r = (r^2+3r)^2$). The multiplicity of τ is $g = \big((v-1)\theta + k\big)/(\theta-\tau) = \big((n_r-1)r + k_r\big)/(r(r+3)) = k_r$, again by direct calculation, and the multiplicity of θ is $n_r - 1 - k_r$. The other Krein condition is automatic when $\lambda = 0$ and $\theta \geq 1$.

(2) By Corollary 7.3 with $\theta = r$ and $m_\tau = k_r$, there is an almost-equidistant set of n_r points in $\mathbb{R}^{k_r}$ on a sphere of squared radius $\frac{1}{2} + \frac{k_r - r}{2n_r r} = \frac{1}{2} + \frac{r^2(r+3)}{2r^3(r+3)^2} = \frac{1}{2} + \frac{1}{2r(r+3)}$. For the expansion, $D = r^3+3r^2+r+1 = r^3(1+x)$ with $x = 3/r + 1/r^2 + 1/r^3$, and $(1+x)^{4/3} = 1 + \frac{4}{3}x + \frac{2}{9}x^2 - \frac{4}{81}x^3 + O(x^4)$ gives $D^{4/3} = r^4 + 4r^3 + \frac{10}{3}r^2 + O(r)$; since $n_r = r^4 + 6r^3 + 9r^2$ and $2D = 2r^3 + 6r^2 + O(r)$, we get $n_r - D^{4/3} - 2D = -\frac{1}{3}r^2 + O(r)$, and $r^2 = D^{2/3} + O(D^{1/3})$.

(3) The first statement is (2), with Theorem 1.1(ii) for the sharpness. For the last one, $n_r/k_r = r(r+3)^2/(r^2+3r+1) > r$, so $f(k_r)/k_r > r$ for every r for which the graph exists, which is incompatible with $f(d) \leq Cd$ for infinitely many r. □

Following Makhnev [22, 10], a strongly regular graph with parameters $(n_r, k_r, 0, \mu_r)$ is denoted Kre(r). Kre(1) and Kre(2) are the Clebsch and Higman–Sims graphs, unique with their parameters, which give the 16- and 100-point sets of Table 1; the pentagon is the member with $r = (\sqrt{5}-1)/2$. Gavrilyuk and Makhnev [12] proved that Kre(3), with parameters $(324, 57, 0, 12)$, does not exist, and Makhnev [22] proved that Kre(4), with parameters $(784, 116, 0, 20)$, does not exist; for $r \geq 5$ the question is open as far as we know [10]. By a theorem of Cameron, Goethals and Seidel [9], a strongly regular graph attaining a Krein condition with equality has strongly regular subconstituents; for Kre(r) the non-neighbours of a vertex form a graph with parameters $((r^2+2r-1)(r^2+3r+1),\ r^3+2r^2,\ 0,\ r^2)$ [10], which is the Petersen graph for $r = 1$, the M_{22} graph for $r = 2$, and the nonexistent $(266, 45, 0, 9)$ [1] for $r = 3$. So the family that would make the $O(d^{4/3})$ bound sharp has no known member beyond $r = 2$, is known to have none at $r = 3, 4$, and the general expectation that there are only finitely many triangle-free strongly regular graphs would exclude all of it. We know of no construction of a large almost-equidistant set outside strongly regular graphs and their direct sums, and of no argument that rules one out.

Proof of Theorem 1.2. This is Proposition 4.1 together with Theorem 1.1(ii): if Kre(r) exists for infinitely many r, then along those r the largest two-distance almost-equidistant set in $\mathbb{R}^{k_r}$ has between $D^{4/3} + 2D - \frac{1}{3}D^{2/3} + O(D^{1/3})$ and $D^{4/3} + 2D + 100D^{2/3}$ points, $D = k_r + 1$. □

Theorem 1.6 applies to any triangle-free strongly regular graph that might still be found. Apart from the Krein family, the feasible parameter sets with $\theta = 3$ are $(162, 21, 0, 3)$, $(176, 25, 0, 4)$, $(210, 33, 0, 6)$ and $(266, 45, 0, 9)$, with $m_\tau = 56, 55, 55, 56$; the last is excluded by Akiyama [1], and a recent preprint of Kovář [19] reports that $(176, 25, 0, 4)$ fails a lattice-theoretic necessary condition, which we have not checked; for $(162, 21, 0, 3)$ and $(210, 33, 0, 6)$ we know of no proof of nonexistence; they would give $f(56) \geq 162$ and $f(55) \geq 210$. A Moore graph of degree 57, parameters $(3250, 57, 0, 1)$, has $\theta = 7$ and $m_\tau = 1520$ and would give only $3250 = 2 \cdot 1520 + 210$.

4.2. **Remark** (Where the known upper bounds stop)**.** The hypothetical sets of Proposition 4.1 sit exactly at the limit of the arguments of [18] and [24]. Polyanskii [24, Theorem 11] proved that an almost-equidistant set contained in a ball of squared radius $\frac{1}{2} + c_0(d+1)^{-2/3}$ with $c_0 < \frac{1}{2}$ has $O(d)/(1-2c_0)$ points; for the set of Proposition 4.1(2) the excess $\frac{1}{2r(r+3)}$ over $\frac{1}{2}$ equals $c_0(d+1)^{-2/3}$ with $c_0 \to \frac{1}{2}$ from below as $r \to \infty$ (numerically $c_0 = 0.40, 0.42, 0.43, 0.44$ for $r = 2, 3, 4, 5$), and the ratio $n_r/k_r \approx r$ grows like $1/(1-2c_0)$. In the same way, the $O(d^{4/3})$ argument of [18] and [24, Theorem 14] bounds the distance of the points from the sphere of radius $1/\sqrt{2}$ through the row sums of the matrix U of Lemma 2.1, and for these sets every row sum is $k_r \cdot \frac{1}{r} = r^2 + 3r + 1 \approx k_r^{2/3}$,

which is the order at which that argument gives nothing. The triangle-free graphs with a positive eigenvalue of multiplicity $n-O(n^{3/4})$ mentioned in Section 5 fit the same template, with eigenvalue $r \approx n^{1/4}$. Proposition 4.1 is a necessary condition for Conjecture 12, not an equivalence: a proof of the conjecture would imply that $\mathrm{Kre}(r)$ does not exist for all large r, but the converse fails, since the conjecture also covers two-distance sets whose graph is not strongly regular and sets with more than two distances.

5. The rank of $A - \lambda_2 I$ for triangle-free graphs

In this section G is a triangle-free graph on the vertex set $[n] = \{1, \dots, n\}$ with adjacency matrix A and eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$, $N(i)$ is the neighbourhood of i, and Δ is the maximum degree.

Proof of Theorem 1.3. Let $\theta = \lambda_2 > 0$, let m be its multiplicity and $r = \operatorname{rank}(A - \theta I) = n - m$.

The case $\lambda_1 = \theta$. All eigenvalues are at most θ, $m = n - r$ of them equal θ, and the remaining r are at least $-\lambda_1 = -\theta$. Since $\operatorname{tr} A = 0$, we get $(n-r)\theta \leq r\theta$, i.e. $n \leq 2r$.

The case $\lambda_1 > \theta$. Now exactly one eigenvalue exceeds θ, so $A - \theta I$ has one positive eigenvalue $\lambda_1 - \theta$, with a unit eigenvector u, a kernel of dimension m, and $r-1$ negative eigenvalues. Put $D = r - 1$. Writing the negative part through its Gram factorisation,

$$A - \theta I = \beta\beta^\top - WW^\top, \qquad \beta = \sqrt{\lambda_1 - \theta}\, u, \tag{7}$$

where W is an $n \times D$ matrix with rows $w_1, \dots, w_n \in \mathbb{R}^D$. Comparing entries, with $\varepsilon_i := \beta_i^2 \geq 0$,

$$\|w_i\|^2 = \theta + \varepsilon_i, \qquad \langle w_i, w_j \rangle = \beta_i\beta_j - A_{ij} \quad (i \neq j), \qquad \sum_{i=1}^{n} \varepsilon_i = \lambda_1 - \theta =: E. \tag{8}$$

Note that $D \geq 1$: if $r = 1$ then $A = \theta I + \beta\beta^\top$ would be positive definite, which a matrix with zero diagonal is not.

Step 1: degrees. Fix a vertex i. Since G is triangle-free, $N(i)$ is an independent set, so by (8) the Gram matrix of $\{w_j : j \in N(i)\}$ is $\theta I + \beta_N \beta_N^\top$, where β_N is the restriction of β to $N(i)$. This matrix is positive definite, so the vectors w_j, $j \in N(i)$, are linearly independent in $\mathbb{R}^D$ and $|N(i)| \leq D$. Hence $\Delta \leq D$, and $E = \lambda_1 - \theta < \lambda_1 \leq \Delta \leq D$.

Step 2: light vertices. Suppose, for a contradiction, that $n \geq 8(\theta+1)D$. Put

$$\delta = \frac{2E}{n} \leq \frac{2D}{n} \leq \frac{1}{4(\theta+1)}, \qquad L = \{i \in [n] : \varepsilon_i \leq \delta\}.$$

Since $\sum_i \varepsilon_i = E$, fewer than $E/\delta = n/2$ vertices have $\varepsilon_i > \delta$, so $|L| > n/2$.

Step 3: light vertices have few light neighbours. Let $i \in L$ and $N' = N(i) \cap L$, and suppose $k := |N'| \geq 1$. By Step 1 the Gram matrix of $\{w_j : j \in N'\}$ is $\Gamma = \theta I + \beta'\beta'^\top$ with β' the restriction of β to N', and by (8) the vector $c \in \mathbb{R}^{N'}$ with $c_j = \langle w_i, w_j \rangle = \beta_i\beta_j - 1$ satisfies $|c_j| \geq 1 - \delta$, because $|\beta_i\beta_j| \leq \sqrt{\varepsilon_i\varepsilon_j} \leq \delta < 1$. The squared norm of the orthogonal projection of w_i onto $\operatorname{span}\{w_j : j \in N'\}$ is $c^\top \Gamma^{-1} c$, and it is at most $\|w_i\|^2 = \theta + \varepsilon_i \leq \theta + \delta$. Since $\beta'\beta'^\top \preceq \|\beta'\|^2 I$ we have $\Gamma \preceq (\theta + \|\beta'\|^2) I$, hence $\Gamma^{-1} \succeq (\theta + \|\beta'\|^2)^{-1} I$ and

$$\frac{k(1-\delta)^2}{\theta + k\delta} \leq \frac{\|c\|^2}{\theta + \|\beta'\|^2} \leq c^\top \Gamma^{-1} c \leq \theta + \delta,$$

using $\|\beta'\|^2 = \sum_{j \in N'} \varepsilon_j \leq k\delta$. Expanding, $k\big((1-\delta)^2 - \delta(\theta+\delta)\big) \leq \theta(\theta+\delta)$, and $(1-\delta)^2 - \delta(\theta+\delta) = 1 - \delta(\theta+2) \geq 1 - \frac{\theta+2}{4(\theta+1)} \geq \frac{1}{2}$. Therefore

$$k \leq 2\theta(\theta+\delta) \leq 2\theta^2 + \frac{\theta}{2(\theta+1)} < 2\theta^2 + \frac{1}{2}.$$

So the maximum degree Δ_L of the induced subgraph $G[L]$ satisfies $\Delta_L < 2\theta^2 + \frac{1}{2}$. As Δ_L is an integer, $\Delta_L = 0$ when $\theta < \frac{1}{2}$, and $\Delta_L/\theta < 2\theta + \frac{1}{2\theta} \leq 2\theta + 1$ when $\theta \geq \frac{1}{2}$.

Step 4: the trace of A_L. Let A_L be the adjacency matrix of $G[L]$. Restricting (7) to L gives $A_L-\theta I=\beta_L\beta_L^\top-W_LW_L^\top$, a matrix of rank at most $D+1$, so θ is an eigenvalue of A_L of multiplicity $m_L\ge|L|-D-1$. The other $R:=|L|-m_L\le D+1$ eigenvalues of A_L are at least $-\Delta_L$. Since $\operatorname{tr}A_L=0$,

$$m_L\theta\le R\,\Delta_L\le(D+1)\Delta_L,\qquad\text{so}\qquad |L|=m_L+R\le(D+1)\Big(1+\frac{\Delta_L}{\theta}\Big).$$

If $\theta<\frac12$ this gives $|L|\le D+1$, and if $\theta\ge\frac12$ it gives $|L|<(D+1)(2\theta+2)$. In both cases $|L|<2(\theta+1)(D+1)\le4(\theta+1)D$, using $D\ge1$. Together with Step 2, $n<2|L|<8(\theta+1)D$, contradicting the assumption of Step 2. Hence $n<8(\theta+1)D=8(\theta+1)(r-1)$.

Finally, in both cases $n<8(\theta+1)r$, since $2r<8(\theta+1)r$. □

Proof of Corollary 1.4. (1) A two-distance almost-equidistant set with distances $a<1$ has diameter 1, and Theorem 9 of [24] states that an almost-equidistant set of diameter 1 in $\mathbb{R}^d$ has at most $2d+4$ points.

(2) If $n\le d+3$ there is nothing to prove. If $n\ge d+4$, Proposition 2.2 gives $\theta=\lambda_2>0$ and $r=\operatorname{rank}(A-\theta I)\le d+2$ for the graph G of the pairs at distance a, and Theorem 1.3 gives $n<8(\theta+1)(r-1)\le8(\theta+1)(d+1)$ if $\lambda_1>\theta$ and $n\le2r\le2d+4<8(\theta+1)(d+1)$ if $\lambda_1=\theta$. Since $\theta+1=a^2/(a^2-1)$, the bound reads $n<8a^2(d+1)/(a^2-1)$.

(3) Larman, Rogers and Seidel [21, Theorem 2] proved that a two-distance set in $\mathbb{R}^d$ with distances $a_1<a_2$ and more than $2d+3$ points satisfies $(a_1/a_2)^2=(k-1)/k$ for an integer k with $2\le k\le\frac12+\sqrt{d/2}$. Here $a_1=1$ and $a_2=a$, so $a^2=k/(k-1)$ and $\theta=1/(a^2-1)=k-1$, and (2) gives $n<8k(d+1)$. □

The first part of Proposition 2.2 is the reduction of Problem 18 to Problem 19 in [24]; Theorem 1.3 bounds the rank in Problem 19 from below by $n/(8(\lambda_2+1))$. Polyanskii notes in the same place, citing [11], that for every n there is a triangle-free graph on n vertices and a positive eigenvalue λ of its adjacency matrix with $\operatorname{rank}(A-\lambda I)=O(n^{3/4})$, which would match the order of magnitude of the upper bound $f(d)=O(d^{4/3})$ if such a λ could be taken to be the second largest eigenvalue of a graph satisfying the positivity condition of Proposition 2.2. Theorem 1.3 shows that in such a graph either $\lambda\ne\lambda_2$ or $\lambda_2+1\ge c'n^{1/4}$, where $c'>0$ depends only on the implied constant in $O(n^{3/4})$; so any such family must have second largest eigenvalue growing at least like $n^{1/4}$.

5.1. **Remark.** For two-distance sets, Theorem 1.3 and the Frobenius norm alone already give the exponent 4/3: Corollary 1.4(2) bounds n by $8(\theta+1)(d+1)$, while $\|U\|_F^2=2|E(G)|/\theta^2\le n(d+1)/\theta^2$ together with Lemma 2.6(1) bounds n by about $(d+1)^{3/2}/\theta$; the two bounds meet at $\theta\approx\frac12(d+1)^{1/3}$ and give $n\le(4+o(1))(d+1)^{4/3}$. The third moment of Lemma 2.5 is what replaces the constant 4 by 1.

6. Non-unit distances at least $a>1$

The proof uses the following estimate from [24].

6.1. **Lemma** ([24, Lemma 6])**.** *There is an absolute constant $c\ge1$ such that if $\{w_0,w_1,\dots,w_k\}$ is an almost-equidistant set in $\mathbb{R}^d$ with $\big|\|w_i\|^2-\frac12\big|\le x$ for $0\le i\le k$, where $x>0$, then*

$$\Big|\sum_{i=1}^{k}\big(\|w_0-w_i\|^2-1\big)\Big|\le c\big(d^{1/2}+d\,x^{1/2}+d\,x\big).$$

Proof of Theorem 1.5. Let $V=\{v_1,\dots,v_n\}$, $g=a^2-1>0$, and let G be the graph on $[n]$ whose edges are the pairs at non-unit distance, so that G is triangle-free, $N(i)$ is a set of points at pairwise unit distance for every i, and $\|v_i-v_j\|^2-1\ge g$ for every edge ij. Let $c\ge1$ be the constant of Lemma 6.1. We show that $n<384c\,d\max\{1,g^{-2}\}$; the theorem then holds with $C=384c$. If $n\le d+1$ or $n<96c\,d$ this is clear, so assume $n\ge d+2$ and $n\ge96c\,d$; in particular G has an edge.

Translate so that $\sum_i v_i = 0$ and put $S = \sum_i \|v_i\|^2$. Then $\sum_{j=1}^n \|v_i - v_j\|^2 = n\|v_i\|^2 + S$, and since the unit pairs contribute 0 to F_i and the term $j = i$ is excluded,

$$F_i := \sum_{j\in N(i)} \big(\|v_i - v_j\|^2 - 1\big) = \sum_{j\neq i} \big(\|v_i - v_j\|^2 - 1\big) = n\|v_i\|^2 + S - n + 1 \geq 0.$$

Let $f = \max\{1, \max_i F_i\}$. Summing over i gives $0 \leq 2nS - n^2 + n \leq nf$, so $-\frac{1}{2} \leq S - \frac{n}{2} \leq \frac{f-1}{2}$, and then $n\|v_i\|^2 - \frac{n}{2} = (F_i - 1) - (S - \frac{n}{2})$ with $-1 \leq F_i - 1 \leq f - 1$ gives $|n\|v_i\|^2 - \frac{n}{2}| \leq \max\{\frac{f+1}{2}, f - \frac{1}{2}\} \leq \frac{3f}{2}$, that is,

$$\Big|\|v_i\|^2 - \frac{1}{2}\Big| \leq x := \frac{3f}{2n} \qquad (1 \leq i \leq n). \tag{9}$$

Applying Lemma 6.1 to $\{v_i\} \cup N(v_i)$ for an index i with $F_i = \max_j F_j$, and using (9) (if $f = 1 > \max_j F_j$ the next inequality holds trivially, as $c\,d^{1/2} \geq 1$),

$$f \leq c\big(d^{1/2} + d\,x^{1/2} + d\,x\big), \qquad x = \frac{3f}{2n}. \tag{10}$$

Since $n \geq 96c\,d$ we have $c\,d\,x = 3cdf/(2n) \leq f/64$, so (10) gives $\frac{63}{64}f \leq c\,d^{1/2} + c\,d\,x^{1/2}$, i.e. $\frac{21}{32}nx \leq c\,d^{1/2} + c\,d\,x^{1/2}$. One of the two terms on the right is at least half the right-hand side, so either $\frac{21}{32}nx \leq 2c\,d^{1/2}$ or $\frac{21}{32}nx \leq 2c\,d\,x^{1/2}$; in the first case $x \leq \frac{64}{21}c\,d^{1/2}/n$ and in the second $x \leq (\frac{64}{21})^2 c^2 d^2/n^2$. Hence

$$x \leq \max\Big\{\frac{64c\,d^{1/2}}{21n},\ \frac{64^2c^2d^2}{21^2n^2}\Big\} \leq \frac{1}{16} \qquad \text{since } n \geq 96c\,d. \tag{11}$$

Since G has an edge ij, $a^2 \leq \|v_i - v_j\|^2 \leq 2\|v_i\|^2 + 2\|v_j\|^2 \leq 2 + 4x \leq \frac{9}{4}$ by (9), so $g \leq \frac{5}{4}$.

Now suppose, for a contradiction, that $n \geq 384c\,d/g^2$. Inserting this into (11) gives

$$x \leq \max\Big\{\frac{64}{21\cdot 384}\,g^2 d^{-1/2},\ \frac{64^2}{21^2\cdot 384^2}\,g^4\Big\} \leq \frac{g^2}{64},$$

using $g^2 \leq \frac{25}{16}$ in the second term.

Degrees. Fix i and let $u_1, \dots, u_k$ be the points of $N(i)$, $k = |N(i)| \geq 1$ (if $k = 0$ there is nothing to show). They are at pairwise unit distance, so $\langle u_s, u_t\rangle = \frac{1}{2}(\|u_s\|^2 + \|u_t\|^2 - 1) \leq x$ for $s \neq t$, and $\langle v_i, u_s\rangle = \frac{1}{2}(\|v_i\|^2 + \|u_s\|^2 - \|v_i - u_s\|^2) \leq \frac{1}{2}(1 + 2x - a^2) = -\frac{1}{2}(g - 2x) < 0$. By the Cauchy–Schwarz inequality applied to v_i and $u_1 + \cdots + u_k$,

$$\frac{k^2(g-2x)^2}{4} \leq \Big(\sum_{s=1}^k \langle v_i, u_s\rangle\Big)^2 \leq \|v_i\|^2 \Big\|\sum_{s=1}^k u_s\Big\|^2 \leq \Big(\frac{1}{2} + x\Big)\Big(k\Big(\frac{1}{2} + x\Big) + k(k-1)x\Big).$$

Dividing by k and rearranging, $k\big(\frac{1}{4}(g-2x)^2 - (\frac{1}{2} + x)x\big) \leq (\frac{1}{2} + x)^2$. With $x \leq g^2/64$ and $g \leq \frac{5}{4}$ we have $2x \leq g/16$, so $\frac{1}{4}(g-2x)^2 \geq \frac{1}{4}(\frac{15}{16})^2 g^2 > \frac{27}{128}g^2$, while $(\frac{1}{2} + x)x \leq \frac{9}{16}\cdot\frac{g^2}{64} = \frac{9}{1024}g^2$ and $(\frac{1}{2} + x)^2 \leq \frac{81}{256}$. Hence

$$k \leq \frac{81/256}{(\frac{27}{128} - \frac{9}{1024})g^2} = \frac{81\cdot 1024}{256\cdot 207\,g^2} < \frac{1.57}{g^2},$$

so $\Delta < 1.57/g^2$.

Trace. Let U be the matrix of Lemma 2.1. Its entries are non-negative, equal to 0 off the edges of G, and at most $2 + 4x - 1 \leq 1 + g^2/16$ on the edges, so every row sum of U is at most $\Delta(1 + g^2/16)$ and, U being symmetric with non-negative entries, every eigenvalue of U is at least $-\Delta(1 + g^2/16)$. By Lemma 2.1, at least $n - d - 2$ eigenvalues of U equal 1 and $\operatorname{tr} U = 0$, so

$$n - d - 2 \leq (d+2)\,\Delta\Big(1 + \frac{g^2}{16}\Big) < (d+2)\,\frac{1.57}{g^2}\Big(1 + \frac{25}{256}\Big) < (d+2)\,\frac{1.73}{g^2},$$

and therefore, using $1 \le \frac{25}{16} g^{-2}$ and $d+2 \le 3d$,

$$n < (d+2)\Big(1 + \frac{1.73}{g^2}\Big) \le (d+2)\,\frac{3.3}{g^2} \le \frac{9.9\,d}{g^2} < \frac{384c\,d}{g^2},$$

contradicting the assumption $n \ge 384c\,d/g^2$. Hence $n < 384c\,d/g^2 \le 384c\,d\max\{1, g^{-2}\}$, and the theorem follows with $C = 384c$. □

The constants in Theorems 1.3 and 1.5 were not optimised. For the sets of Theorem 1.6 the ratio n/d is at most $100/22 < 5$ for the known graphs, while Corollary 1.4 only gives $n < 24(d+1)$ when $\theta = 2$.

7. Constructions from triangle-free graphs

7.1. **Spherical almost-equidistant sets and direct sums.** For a set of points on a sphere of radius r centred at the origin, two points p, q satisfy $\|p-q\|^2 = 2r^2 - 2\langle p, q\rangle$, so

$$\|p-q\| = 1 \iff \langle p, q\rangle = r^2 - \tfrac{1}{2}. \tag{12}$$

It is convenient to normalise: for $c \in (-1, 1)$ let $h(d, c)$ be the largest size of a set of unit vectors in $\mathbb{R}^d$ such that among any three of them two have inner product exactly c. Scaling by $r = (2(1-c))^{-1/2}$ shows that $h(d, c)$ is the largest size of an almost-equidistant set on the sphere of radius r in $\mathbb{R}^d$, where $c = 1 - 1/(2r^2)$. Rosenfeld [25] proved $h(d, 0) = 2d$, and Bezdek and Lángi [4] proved $h(d, c) \le 2d+2$ for $c \le 0$, with equality for $c = -1/d$. Bachoc, Bekker, Moustrou and de Oliveira Filho [2] showed that equality holds only for $c = -1/d$, and gave a semidefinite programming upper bound on $h(d, c)$ for $c \le 0$ [2, Theorem 1.1].

7.1. **Lemma** (Direct sum)**.** *Let $P \subset \mathbb{R}^{d_1}$ and $Q \subset \mathbb{R}^{d_2}$ be almost-equidistant sets such that $\|p\| = r_1$ for all $p \in P$, $\|q\| = r_2$ for all $q \in Q$, and $r_1^2 + r_2^2 = 1$. Then $(P \times \{0\}) \cup (\{0\} \times Q)$ is an almost-equidistant set of $|P| + |Q|$ points in $\mathbb{R}^{d_1+d_2}$.*

Proof. Write $P' = P \times \{0\}$ and $Q' = \{0\} \times Q$. If $r_1 = 0$ or $r_2 = 0$ then P or Q is a single point at distance 1 from every point of the other set, and the statement is trivial, so assume $r_1, r_2 > 0$; then $P' \cap Q' = \emptyset$ and $|P' \cup Q'| = |P| + |Q|$. For $p \in P$ and $q \in Q$ we have $\|(p,0) - (0,q)\|^2 = \|p\|^2 + \|q\|^2 = r_1^2 + r_2^2 = 1$. Let T be a set of three points of $P' \cup Q'$. If T meets both P' and Q', then T contains a pair $\{(p,0),(0,q)\}$, which is at distance 1. Otherwise $T \subset P'$ or $T \subset Q'$, and T contains a pair at distance 1 because P and Q are almost-equidistant. □

In the normalised language, if $1/(1-c_1) + 1/(1-c_2) = 2$ then $f(d_1+d_2) \ge h(d_1, c_1) + h(d_2, c_2)$. Solving, $c_2 = c_1/(2c_1 - 1)$; in particular exactly one of c_1, c_2 is positive unless $c_1 = c_2 = 0$. The negative summand is controlled by the Bezdek–Lángi bound; the next lemma records what is attained by two cliques.

7.2. **Lemma** (Two cliques)**.** *Let $j \ge 1$ and $c \in (-1, 0)$.*

1. *If $c > -1/(j-1)$ (no condition if $j = 1$), then $h(j, c) \ge 2j$.*
2. *If $c = -1/j$, then $h(j, c) \ge 2j + 2$.*

Proof. (1) The $j \times j$ matrix $G = (1-c)I + cJ$ has eigenvalues $1-c$ with multiplicity $j-1$ and $1+(j-1)c$ with multiplicity 1; both are positive when $c > -1/(j-1)$, so G is positive definite and is the Gram matrix of j unit vectors $u_1, \dots, u_j$ in $\mathbb{R}^j$ with $\langle u_a, u_b\rangle = c$ for $a \ne b$. Put $S = \{u_1, \dots, u_j\}$. The set $S \cup (-S)$ has $2j$ distinct elements: $u_a = -u_b$ would give $\langle u_a, u_b\rangle = -1 < c$ if $a \ne b$ and $\|u_a\| = 0$ if $a = b$. Any three elements of $S \cup (-S)$ contain two from S or two from $-S$, and those have inner product c. Hence $h(j, c) \ge 2j$.

(2) For $c = -1/j$ the $(j+1) \times (j+1)$ matrix $(1 + 1/j)I - (1/j)J$ is positive semidefinite of rank j (its eigenvalues are $1 + 1/j$, with multiplicity j, and 0), so it is the Gram matrix of $j+1$ unit vectors in $\mathbb{R}^j$ with pairwise inner products $-1/j$: the vertices of a regular simplex centred at the

origin. As in (1), the union of this simplex and its antipodal image is a set of $2j+2$ unit vectors in $\mathbb{R}^j$ in which any three contain a pair with inner product $-1/j$. □

For $c=-1/3$, $j=3$, the set in (2) is the vertex set of a cube; this is the 8-point set used in [3] for $f(8)\geq 24$.

7.2. **The spectral embedding.**

Proof of Theorem 1.6. Let A be the adjacency matrix of H, J the all-ones matrix and $\mathbf{1}$ the all-ones vector. Since H is k-regular and connected, $A\mathbf{1}=k\mathbf{1}$ and k is a simple eigenvalue, so $\lambda_2<k$. Let $u_1=\mathbf{1}/\sqrt{n}, u_2,\dots,u_n$ be an orthonormal eigenbasis of A with eigenvalues $k=\lambda_1>\lambda_2\geq\lambda_3\geq\cdots\geq\lambda_n$. Define

$$M=\lambda_2 I-A+\frac{k-\lambda_2}{n}J.$$

Then $Mu_1=\lambda_2u_1-ku_1+(k-\lambda_2)u_1=0$, and for $i\geq 2$, $Ju_i=0$ (as $u_i\perp\mathbf{1}$) gives $Mu_i=(\lambda_2-\lambda_i)u_i$ with $\lambda_2-\lambda_i\geq 0$. Hence M is symmetric positive semidefinite, and its kernel is $\operatorname{span}(\mathbf{1})\oplus\ker(A-\lambda_2 I)$, of dimension $1+m$. Therefore $\operatorname{rank}M=n-1-m$, and there are vectors $p_1,\dots,p_n\in\mathbb{R}^{n-1-m}$ with $\langle p_i,p_j\rangle=\rho M_{ij}$ for all i,j, where $\rho=1/(2\lambda_2)>0$ (take the rows of $\sqrt{\rho}\,M^{1/2}$ restricted to the range of M).

The entries of M are $M_{ii}=\lambda_2+(k-\lambda_2)/n$, $M_{ij}=(k-\lambda_2)/n$ if $i\neq j$ are non-adjacent, and $M_{ij}=-1+(k-\lambda_2)/n$ if i,j are adjacent. Consequently

$$\|p_i-p_j\|^2=\rho(M_{ii}+M_{jj}-2M_{ij})=\begin{cases}2\rho\lambda_2=1, & i\not\sim j,\\ 2\rho(\lambda_2+1)=1+1/\lambda_2, & i\sim j,\end{cases}$$

and $\|p_i\|^2=\rho M_{ii}=\frac{1}{2}+\frac{k-\lambda_2}{2n\lambda_2}=r^2$. The points are distinct since all distances are at least 1. If i,j,l are three vertices, they are not all pairwise adjacent because H is triangle-free, so some pair among p_i,p_j,p_l is at distance 1. Thus $\{p_1,\dots,p_n\}$ is almost-equidistant in $\mathbb{R}^{n-1-m}$, lies on the sphere of radius r, and realises only the distances 1 and $\sqrt{1+1/\lambda_2}$. □

Write $d=n-1-m$. The excess of the set over the general bound $2d+4$ is $n-2d-4=2m-n-2$, so Theorem 1.6 beats $2d+4$ exactly when λ_2 has multiplicity larger than $n/2+1$. For bipartite graphs the spectrum is symmetric and $m\leq(n-2)/2$, so bipartite graphs never help; this is consistent with $h(d,0)=2d$.

7.3. **Corollary.** *Let H be a strongly regular graph with parameters $(v,k,0,\mu)$ (so H is triangle-free) and eigenvalues $k>\theta>\tau$ with multiplicities $1,m_\theta,m_\tau$. Then Theorem 1.6 applies with $n=v$, $\lambda_2=\theta$, $m=m_\theta$, and yields an almost-equidistant set of v points on a sphere of radius r in $\mathbb{R}^{m_\tau}$, where*

$$r^2=\frac{1}{2}+\frac{k-\theta}{2v\theta},\qquad c:=1-\frac{1}{2r^2}=\frac{k-\theta}{v\theta+k-\theta}.$$

The direct sum with the set of Lemma 7.2 then gives, with $q:=v\theta/(k-\theta)-1$,

$$f(m_\tau+j)\geq v+2j\quad(1\leq j<q),\qquad f(m_\tau+q)\geq v+2q+2\quad\text{if } q\in\mathbb{Z}.$$

Proof. A strongly regular graph is connected with θ its second largest eigenvalue, $\theta>0$ since $\theta\tau=\mu-k<0$ and $\tau<0$; and $v-1-m_\theta=m_\tau$. The first statement is Theorem 1.6. For the second, the partner sphere has squared radius $1-r^2$, i.e. normalised inner product $c_2=c/(2c-1)=-(k-\theta)/(v\theta-k+\theta)=-1/q$. Lemma 7.2 gives $h(j,-1/q)\geq 2j$ for $j<q$ (as $-1/q>-1/(j-1)$) and $h(q,-1/q)\geq 2q+2$ when q is an integer, and Lemma 7.1 combines the two sets. □

7.4. **Remark** (The third moment and the Krein condition)**.** In the proof of Theorem 1.6 the matrix M acts as multiplication by $\theta-\tau$ on the τ-eigenspace and as 0 on its orthogonal complement, so for a strongly regular graph $M=(\theta-\tau)E_\tau$, where E_τ is the orthogonal projection onto the τ-eigenspace, and the Gram matrix of P is $\rho(\theta-\tau)E_\tau$. Let $q^\tau_{\tau\tau}$ be the Krein parameter defined by

$E_\tau \circ E_\tau = \frac{1}{n}\sum_\varepsilon q^\varepsilon_{\tau\tau} E_\varepsilon$, the sum over the three primitive idempotents. Since $\mathbf{1}^\top (E_\varepsilon \circ E_\tau)\mathbf{1} = \operatorname{tr}(E_\varepsilon E_\tau)$ vanishes unless $\varepsilon = \tau$, where it equals m_τ,

$$\sum_{i,j} \langle p_i, p_j\rangle^3 = \rho^3(\theta-\tau)^3\, \mathbf{1}^\top (E_\tau \circ E_\tau \circ E_\tau)\mathbf{1} = \rho^3(\theta-\tau)^3\, \frac{q^\tau_{\tau\tau} m_\tau}{n}.$$

So for these sets the inequality $\sum_{i,j}\langle p_i, p_j\rangle^3 \geq 0$ behind Lemma 2.5 is the Krein condition $q^\tau_{\tau\tau} \geq 0$, which in the parameters reads $(\tau+1)(k+\tau+2\theta\tau) \leq (k+\tau)(\theta+1)^2$, and it holds with equality exactly for the graphs of Proposition 4.1. The script of Appendix A confirms that the third moment vanishes for the Clebsch and Higman–Sims sets.

Table 1 lists the data for the six known triangle-free strongly regular graphs other than the pentagon. (For the pentagon, $\theta = (\sqrt{5}-1)/2$ has multiplicity 2 and the theorem gives 5 points in $\mathbb{R}^2$, which is below the Moser spindle.) The Clebsch row is exactly the Larman–Rogers set and, with its partner cube, the 24-point set of [3]; these two rows serve as checks of the method against the literature.

graph	(v,k,λ,μ)	θ	τ	m_τ	r^2	c	q	bounds
Petersen	$(10,3,0,1)$	1	-2	4	3/5	1/6	4	$f(4)\geq 10$, $f(8)\geq 20$
Clebsch	$(16,5,0,2)$	1	-3	5	5/8	1/5	3	$f(5)\geq 16$, $f(8)\geq 24$
Hoffman–Singleton	$(50,7,0,1)$	2	-3	21	21/40	1/21	19	$f(21)\geq 50$, $f(40)\geq 90$
Gewirtz	$(56,10,0,2)$	2	-4	20	15/28	1/15	13	$f(20)\geq 56$, $f(33)\geq 84$
M_{22}	$(77,16,0,4)$	2	-6	21	6/11	1/12	10	$f(21)\geq 77$, $f(31)\geq 99$
Higman–Sims	$(100,22,0,6)$	2	-8	22	11/20	1/11	9	$f(22)\geq 100$, $f(31)\geq 120$

TABLE 1. The six triangle-free strongly regular graphs and the almost-equidistant sets of Corollary 7.3. The last column gives the standalone bound $f(m_\tau)\geq v$ and the direct sum with a double q-simplex.

For the Higman–Sims graph the intermediate values $1 \leq j \leq 8$ of Corollary 7.3 give $f(22+j) \geq 100+2j$, which is the middle part of Theorem 1.7. The bound $f(10)\geq 26$ uses the other eigenspace of the Petersen graph. The orthogonal projection onto its θ-eigenspace ($\theta = 1$, multiplicity 5) is $E_\theta = (4I + 2A - J)/6$, with diagonal entries $1/2$ and off-diagonal entries $1/6$ (adjacent) and $-1/6$ (non-adjacent); it is positive semidefinite of rank 5, so $\frac{3}{4}E_\theta$ is the Gram matrix of 10 points $p_1,\dots,p_{10}\in\mathbb{R}^5$ with $\|p_i\|^2 = 3/8$ and $\|p_i - p_j\|^2 = \frac{3}{4}(1-2(E_\theta)_{ij})$, which equals 1 for non-adjacent and $1/2$ for adjacent pairs. The non-adjacent pairs of the Petersen graph form the triangular graph $T(5)$, whose complement is triangle-free, so this 10-point set is almost-equidistant. Since $3/8 + 5/8 = 1$, Lemma 7.1 combines it with the 16-point Clebsch set (squared radius $5/8$) in an orthogonal $\mathbb{R}^5$, giving 26 points in $\mathbb{R}^{10}$.

7.3. **Extension points from Delsarte-tight cocliques.** Let H be a strongly regular graph with parameters $(v,k,0,\mu)$ and eigenvalues $k > \theta > \tau$, and let $P = \{p_1,\dots,p_v\} \subset \mathbb{R}^{m_\tau}$ be the set of Corollary 7.3. For a coclique C of H, the Delsarte–Hoffman bound states $|C| \leq v(-\tau)/(k-\tau)$, with equality if and only if every vertex outside C has exactly $-\tau$ neighbours in C (see [14, Section 9.6] or [8]). We call such cocliques *tight*.

7.5. **Lemma.** *With the notation above, let $D \geq m_\tau$, regard P as a subset of $\mathbb{R}^{m_\tau} \subseteq \mathbb{R}^D$, let $x \in \mathbb{R}^D \setminus P$ and $C = \{i : \|x - p_i\| \neq 1\}$.*

(1) *If $P\cup\{x\}$ is almost-equidistant, then either $C = \emptyset$ (and then x is orthogonal to* span P *with $\|x\|^2 = 1 - r^2$), or C is a tight coclique of H.*
(2) *Conversely, for every tight coclique C there are exactly two points $x^+, x^- \in \mathbb{R}^{m_\tau}$ with $\{i : \|x^\pm - p_i\| \neq 1\} \subseteq C$, and both $P\cup\{x^+\}$ and $P\cup\{x^-\}$ are almost-equidistant sets of $v+1$ points in $\mathbb{R}^{m_\tau}$.*

Proof. We use the realisation from the proof of Theorem 1.6: $p_i = \sqrt{\rho}\, M^{1/2} e_i$ with $\rho = 1/(2\theta)$ and $M = \theta I - A + \frac{k-\theta}{v} J$. The range W of M is the τ-eigenspace of A (the θ-eigenspace and $\mathbf{1}$ span the kernel of M, and M acts on the τ-eigenspace as multiplication by $\theta - \tau$); P spans W, and we identify $\mathbb{R}^{m_\tau}$ with W. Let $U = \{1, \dots, v\} \setminus C$.

(1) If $i, j \in C$ were adjacent, the triple $\{x, p_i, p_j\}$ would contain no pair at distance 1 (adjacent pairs are at distance $\sqrt{1 + 1/\theta}$); so C is a coclique. Write $x = x' + x''$ with $x' \in W$ and $x'' \perp W$. For $i \in U$, the condition $\|x - p_i\|^2 = 1$ reads $\|x\|^2 - 2\langle x', p_i\rangle + r^2 = 1$, so

$$\langle x', p_i\rangle = \alpha := \tfrac{1}{2}(\|x\|^2 + r^2 - 1) \qquad \text{for all } i \in U.$$

Put $z = \sqrt{\rho}\, M^{1/2} x' \in W$; then $\langle x', p_i\rangle = \langle x', \sqrt{\rho}\, M^{1/2} e_i\rangle = z_i$, so z is a τ-eigenvector of A that equals α on U.

Case $\alpha = 0$. Then z vanishes on U. For $i \in C$ all neighbours of i lie in U (as C is a coclique), hence $\tau z_i = (Az)_i = \sum_{j \sim i} z_j = 0$ and $z_i = 0$. So $z = 0$, hence $x' = 0$ (as $M^{1/2}$ is injective on W), hence $\langle x, p_i\rangle = 0$ for all i and $\|x - p_i\|^2 = \|x\|^2 + r^2$ is the same for all i. Therefore $C = \emptyset$ or $C = \{1, \dots, v\}$; the latter is not a coclique since H has edges. Thus $C = \emptyset$, $x = x'' \perp W$ and $\|x\|^2 = 1 - r^2$.

Case $\alpha \neq 0$. Let $y = z/\alpha$, a τ-eigenvector equal to 1 on U. For $i \in C$, all neighbours of i are in U, so $\tau y_i = (Ay)_i = k$ and $y_i = k/\tau$. For $i \in U$ let $t_i = |N(i) \cap C|$; then $\tau = (Ay)_i = (k - t_i) + t_i\, k/\tau$, i.e. $t_i(k/\tau - 1) = \tau - k$, i.e. $t_i = -\tau$. In particular $C \neq \emptyset$ (as $\tau \neq 0$). Counting the edges between C and U gives $k|C| = -\tau|U| = -\tau(v - |C|)$, so $|C| = v(-\tau)/(k - \tau)$ and C is tight.

(2) Let C be a tight coclique and $y = \mathbf{1}_U + \frac{k}{\tau}\mathbf{1}_C$. For $i \in C$, $(Ay)_i = k = \tau y_i$; for $i \in U$, $(Ay)_i = (k + \tau) \cdot 1 + (-\tau) \cdot k/\tau = \tau = \tau y_i$. So $y \in W$ and $M^{1/2} y = \sqrt{\theta - \tau}\, y$. For real γ put

$$x_\gamma = \frac{\gamma}{\sqrt{\rho(\theta - \tau)}}\, y \in W, \qquad \text{so that} \qquad \langle x_\gamma, p_i\rangle = \sqrt{\rho}\, (M^{1/2} x_\gamma)_i = \gamma\, y_i.$$

For $i \in U$ the condition $\|x_\gamma - p_i\|^2 = 1$ becomes $\|x_\gamma\|^2 - 2\gamma + r^2 = 1$, i.e.

$$\frac{\|y\|^2}{\rho(\theta - \tau)}\, \gamma^2 - 2\gamma + (r^2 - 1) = 0. \tag{13}$$

Since $r^2 < 1$, the product of the roots of (13) is negative, so it has two real roots $\gamma^+ > 0 > \gamma^-$; put $x^\pm = x_{\gamma^\pm}$. Then $\|x^\pm - p_i\| = 1$ for all $i \in U$. Any three points of $P \cup \{x^\pm\}$ contain a pair at distance 1: if none of them is $x^\pm$ this holds because P is almost-equidistant; if the triple is $\{x^\pm, p_i, p_j\}$ with $i \in U$ or $j \in U$, the corresponding pair is at distance 1; and if $i, j \in C$ then $i \not\sim j$ and $\|p_i - p_j\| = 1$. Next, $x^\pm \notin P$: if $x^\pm = p_i$, then $\|x^\pm - p_i\| = 0 \neq 1$ and $\|x^\pm - p_j\| = \sqrt{1 + 1/\theta} \neq 1$ for $j \in N(i)$, so $\{i\} \cup N(i) \subseteq \{j : \|x^\pm - p_j\| \neq 1\} \subseteq C$; but C is a coclique and $N(i) \neq \emptyset$, a contradiction. Finally, if $x \in W$ satisfies $\{i : \|x - p_i\| \neq 1\} \subseteq C$, then part (1) applied with $x'' = 0$ shows $x = x_\alpha$ with α a root of (13) (indeed $z = \alpha y$ because $z - \alpha y$ is a τ-eigenvector vanishing on U, hence zero by the argument of the case $\alpha = 0$), so $x \in \{x^+, x^-\}$. □

7.6. **Remark.** The proof of the first part is written with the degenerate case $x \perp W$ spelled out because it is exactly the mechanism behind the bounds $f(6) \geq 18$ and $f(7) \geq 20$ of [3]: a point orthogonal to W at distance $\sqrt{1 - r^2}$ from the origin is at distance 1 from all of P, and Lemma 7.1 is the systematic version of this.

Two extension points x_1, x_2 coming from tight cocliques C_1, C_2 can be added simultaneously provided every triple $\{x_1, x_2, p_i\}$ contains a unit pair, i.e. provided $C_1 \cap C_2 = \emptyset$ or $\|x_1 - x_2\| = 1$ (and $x_1 \neq x_2$); three or more extension points additionally need a unit pair among themselves. Which of these happen is a finite exact computation:

7.7. **Proposition.** (1) *The Petersen graph has* 5 *tight cocliques (the* 4*-sets of pairs through a fixed element of* $\{1, \dots, 5\}$*). For two of them,* $C_1 \neq C_2$*, the points* x_1^+ *and* x_2^- *are at distance* 1*; hence* $f(4) \geq 12$.

(2) *The Hoffman–Singleton graph has* 100 *tight cocliques of size* 15*, among which there are disjoint pairs; hence* $f(21) \geq 52$.
(3) *The Gewirtz graph has* 42 *tight cocliques of size* 16*, among which there are disjoint pairs; hence* $f(20) \geq 58$.
(4) *The* M_{22} *graph has* 22 *tight cocliques of size* 21 *(the blocks of* $S(3,6,22)$ *through a point), any two of which meet; hence* $f(21) \geq 78$.
(5) *The Clebsch and Higman–Sims graphs have no tight cocliques (*$v(-\tau)/(k-\tau) = 6 > \alpha = 5$ *and* $80/3 \notin \mathbb{Z}$ *respectively). Hence, by Lemma 7.5(1), every almost-equidistant set containing their set* P*, in any dimension, is of the form of Lemma 7.1:* P *together with an almost-equidistant set lying on the sphere of squared radius* $1 - r^2$ *in the orthogonal complement of* $\operatorname{span} P$*. In particular no point can be added to* P *inside* $\mathbb{R}^{m_\tau}$.

In cases (2)–(4), no larger family of extension points can be added to P *inside* $\mathbb{R}^{m_\tau}$.

Proof. The enumeration of tight cocliques and the distance computations are performed exactly by the script in Appendix A (the distances $\|x_1 - x_2\|^2$ lie in the real quadratic field generated by the discriminant of (13), and the script computes in that field). The almost-equidistant property of each resulting set is checked triple by triple. Item (1) recovers the known value $f(4) \geq 12$ [3]. For the last sentence, the script forms the $2|\mathcal{C}|$ candidate points $x_C^{\pm}$, where $\mathcal{C}$ is the set of tight cocliques, computes all their pairwise distances exactly in the same quadratic field, and finds that no three candidates can be added together, and for M_{22} no two. By Lemma 7.5 every extension point in $\mathbb{R}^{m_\tau}$ is one of the $x_C^{\pm}$, so this check is exhaustive. □

Extension points lie off the sphere of P and therefore cannot be combined with Lemma 7.1; this is why $f(23) \geq 102$ uses the Higman–Sims set with two antipodal points rather than the 78-point set. If one extra dimension is allowed, extension points can be made pairwise at distance 1, and the three largest graphs form a tower of this kind. The Gewirtz graph is the subgraph of the M_{22} graph induced on the 56 blocks avoiding a fixed point; each of the other 21 blocks is adjacent exactly to the 16 blocks disjoint from it, which form a tight coclique of the Gewirtz graph, and these 21 blocks are pairwise non-adjacent. Likewise the M_{22} graph is the subgraph of the Higman–Sims graph induced on the 77 non-neighbours of a vertex, and each of the 22 neighbours of that vertex is adjacent exactly to the 21 blocks through it, a tight coclique of the M_{22} graph. Since $\theta = 2$ for all three graphs, unit distance corresponds to non-adjacency and the other distance is $\sqrt{3/2}$ in each case. So the M_{22} set is the Gewirtz set together with 21 extension points in one extra dimension, and the Higman–Sims set is the M_{22} set together with 22 extension points in one extra dimension and one further point.

7.4. **Proof of Theorem 1.7.**

Proof of Theorem 1.7. $f(10) \geq 26$ was established after Table 1; $f(20) \geq 58$, $f(21) \geq 78$ are Proposition 7.7(3),(4); $f(22) \geq 100$ is the Higman–Sims row of Table 1; $f(22+j) \geq 100 + 2j$ for $1 \leq j \leq 8$ and $f(31) \geq 120$ are Corollary 7.3 for the Higman–Sims graph ($q = 9$). Monotonicity $f(d+1) \geq f(d)$ is clear (embed $\mathbb{R}^d \subset \mathbb{R}^{d+1}$), and $120 > 2d + 4$ for $d \leq 57$. Each of these point sets (and the checks $f(4) \geq 12$, $f(5) \geq 16$, $f(8) \geq 24$ against the literature, twenty-five sets in all) is verified exactly by the script in Appendix A, whose printed summary includes

d	4	5	8	10	20	21	22	23	24	25	26	27	28	29	30	31
n	12	16	24	26	58	78	100	102	104	106	108	110	112	114	116	120

□

8. Remarks and questions

8.1. **Remark** (On the ratio $f(d)/d$)**.** The best ratios $f(d)/d$ now known are $100/22 \approx 4.55$ at $d = 22$ and $120/31 \approx 3.87$ at $d = 31$, compared with $16/5 = 3.2$ at $d = 5$ previously. Polyanskii

[24] showed that almost-equidistant sets contained in a ball of radius $1/\sqrt{2}+cd^{-2/3}$, $c<1/2$, have $O(d)$ points; the sets of Theorem 1.6 live on spheres of radius $r>1/\sqrt{2}$ with $r^2-\frac{1}{2}=\frac{k-\lambda_2}{2n\lambda_2}$, e.g. $r^2=11/20$ for Higman–Sims, well outside that regime. Theorem 1.5 covers them from the other side, through their second distance $\sqrt{3/2}$, and Theorem 1.1 covers them because they are short-free.

8.2. **Remark** (What the constructions cannot do)**.** A triangle-free strongly regular graph gives v points in dimension m_τ, which beats $2d+4$ exactly when $2m_\theta>v+2$, and $v+2q+2$ points in dimension $m_\tau+q$ when q is an integer (Corollary 7.3), which beats $2d+4$ exactly when $2m_\theta>v$. Five of the seven known triangle-free strongly regular graphs, all but the pentagon and the Petersen graph, have $2m_\theta>v+2$; whether infinitely many such graphs exist is a well-known open problem [8]. By Corollary 1.4, however, no sequence of such graphs with bounded θ can give $f(d)\ge Cd$ with $C>8(\theta+1)$, and no sequence with bounded θ can give superlinear growth. The direct sums of Lemma 7.1 are limited for a different reason: a spherical almost-equidistant set of squared radius $\rho<\frac{1}{2}$ contains no unit simplex with more than $1+2\rho/(1-2\rho)$ vertices, so by Ramsey's theorem its size is bounded in terms of ρ alone, and the partner of a set of squared radius above $\frac{1}{2}$ has bounded size in every dimension. This is why Theorem 1.7 stops at $d=57$.

8.3. **Remark** (The slack of Section 3 on the known configurations)**.** Table 2 evaluates, exactly, the quantities entering the proof of Theorem 1.1 on the six strongly regular sets of Table 1 and on the hypothetical sets of Kre(r), $r\le 7$ (for which everything is a closed formula: $\eta=1/r$ on all far pairs, $F=k_r/r$, $X=\frac{1}{2}r(r+3)$, $E_2=n_rk_r/r^2$). For these sets all η_{ij} on far pairs are equal, so every Cauchy–Schwarz, Hölder and power-mean step in Section 3 is an equality, the only inequality in Lemma 2.3(3) being $k_i\le D$ (with $k_i=D-1$). The term dropped in (1) is $12\sum y_{ij}^2\eta_{ij}=3n_rF/(r^2(r+3)^2)=O(nD^{-2/3})$, and the bound on L in Lemma 2.5 loses the factor 1.153 (the sets have $\gamma=\frac{1}{2}$ in that proof). The one step that is not tight to first order is the rank bound of Lemma 2.6(1), $E_2\ge n\mu$: the ratio $E_2/(n\mu)$ is $1+1/r+O(r^{-2})$ for Kre(r), because the eigenvalues of U other than 1 are k_r/r (once) and $-(r+2)$ (k_r times) rather than all equal. This ratio alone accounts for the difference between the second-order coefficient 3 that the chain with Lemma 2.6(1) produces (Remark 3.3) and the coefficient 2 of the family; Lemma 2.6(2) recovers the missing $\lambda_0^2=(k_r/r)^2\approx n$ and gives 2. The last column of the table shows how far the chain is from the actual sizes in small dimensions, where the term $6D$ dominates $D^{4/3}$.

set	n	D	$D^{4/3}$	$(n-D^{4/3})/D$	$E_2/(n\mu)$	dropped/n	$n_{\max}$
Petersen	10	5	8.5	0.29	4.50	0.36	31
Clebsch = Kre(1)	16	6	10.9	0.85	3.89	0.94	38
Hoffman–Singleton	50	22	61.6	−0.53	1.49	0.03	154
Gewirtz	56	21	57.9	−0.09	1.62	0.08	147
M_{22}	77	22	61.6	0.70	1.70	0.20	154
Higman–Sims = Kre(2)	100	23	65.4	1.50	1.74	0.33	162
Kre(3) (does not exist)	324	58	224.5	1.72	1.41	0.18	455
Kre(5)	1600	206	1216.6	1.86	1.22	0.08	1968
Kre(7)	4900	498	3947.4	1.91	1.15	0.04	5644

TABLE 2. The quantities of Section 3 on the known and hypothetical strongly regular configurations: "dropped" is the term $12\sum y_{ij}^2\eta_{ij}$ of (1), $E_2/(n\mu)$ the slack of Lemma 2.6(1), and $n_{\max}$ the largest n not excluded by Proposition 3.1 with Lemma 2.6(2) in dimension $d=D-1$ (computed numerically from the exact chain, see `chain_bounds.py`).

8.4. **Remark** (Scope of the searches)**.** The following statements are the outcome of computer searches, and we do not claim optimality beyond them. (i) A search over all triangle-free Cayley graphs on cyclic groups of order at most 34 and over random maximal sum-free connection sets in $\mathbb{Z}_2^k$, $k \le 7$, found no triangle-free Cayley graph on n vertices with $\lambda_2 > 0$ of multiplicity larger than $n/2+1$ other than the Clebsch graph (the folded 5-cube); bipartite and $2d$-type examples appear in abundance. (ii) For the partner spheres relevant to Table 1 we enumerated, for $(d,c) \in \{(4,-\frac{1}{3}),(6,-\frac{1}{3}),(7,-\frac{1}{3}),(3,-\frac{1}{9}),(4,-\frac{1}{9})\}$, all triangle-free graphs on $2d+1$ and $2d+2$ vertices (for $c=-\frac{1}{3}$ and $d \ge 6$, on 12 and 13 vertices, the most that $R(3,5)=14$ allows) whose complement has clique number at most the maximum size of a c-clique, and attempted to realise the complement of each edge-maximal one as unit vectors in $\mathbb{R}^d$ with inner product c on its edges by quasi-Newton minimisation from 20–40 random starts; every attempt stalled at residual at least $3 \cdot 10^{-3}$. Within this search we therefore found no 9-point set for $(4,-\frac{1}{3})$, no 7-point set for $(3,-\frac{1}{9})$, no 9-point set for $(4,-\frac{1}{9})$, and no 11-point set for $(6,-\frac{1}{3})$, i.e. nothing beyond the two-clique sets of Lemma 7.2; we do not claim that these values of h are exact. One of them is settled by the bound of [2, Theorem 1.1], which gives $h(3,-\frac{1}{9}) \le 6$, so $h(3,-\frac{1}{9}) = 6$. At $(4,-\frac{1}{3})$ and $(4,-\frac{1}{9})$ the same bound gives $h \le 9$, so a 9-point set is not excluded; it would give $f(9) \ge 25$ and $f(26) \ge 109$ respectively. This is why Theorem 1.7 has $100+2j$ rather than $100+2j+1$ for $23 \le d \le 30$, and why $f(9)$ and $f(11)$ are not improved. (iii) A simulated-annealing search for large almost-equidistant subsets of the root systems D_4–D_8, E_6, E_7, E_8, of $\{\pm1\}^k$ ($k \le 7$) and of $\{0,\pm1\}^5$, for every squared distance occurring, found nothing exceeding the previously known bounds in dimensions 4–9 (the largest were 17 points of affine dimension 6 in E_7 and 21 points of affine dimension 8 in E_8, both below 18 and 24). The code for these searches is available from the authors.

8.5. **Question.** Is $f(d)$ bounded by Cd for an absolute constant C? By Theorem 1.5, a sequence of almost-equidistant sets with $|V_d|/d \to \infty$ must contain non-unit pairs at distance less than $1+\varepsilon$ for every $\varepsilon > 0$ (possibly below 1); by Corollary 1.4, a two-distance such sequence must have second distance $\sqrt{k/(k-1)}$ with $k \to \infty$. The constructions here give $C \ge 100/22$, and by Proposition 4.1 a positive answer implies that the parameter sets $(r^2(r+3)^2, r(r^2+3r+1), 0, r(r+1))$ are realised for only finitely many r. A more modest question is whether Theorem 1.1 extends to all almost-equidistant sets, i.e. whether $f(d) \le (1+o(1))d^{4/3}$; the obstacle is the pairs at distance below 1 (Remark 3.4).

8.6. **Question.** In Theorem 1.3, can the factor λ_2+1 be replaced by a smaller function of λ_2? For graphs arising from two-distance almost-equidistant sets, Corollary 1.4(3) and $f(d) = O(d^{4/3})$ [18] give $n = O(d^{4/3})$ while λ_2 can be as large as $\sqrt{d/2}$, so for those graphs a factor of order $\lambda_2^{2/3}$ would be consistent with everything that is known.

8.7. **Question.** Is the second-order term in Theorem 1.1(ii) sharp without any hypothesis on strongly regular graphs, that is, are there short-free almost-equidistant sets with $(d+1)^{4/3} + (2 - o(1))(d+1)$ points in infinitely many dimensions? The only candidates we know are the sets of Proposition 4.1.

8.8. **Question.** Is there a 101-point almost-equidistant set in $\mathbb{R}^{22}$? Lemma 7.5 shows that none is obtained by adding a point to the Higman–Sims set.

Appendix A. Exact verification script

The script below is also provided as an ancillary file, `verify_certificate.py`; a second ancillary file, `chain_bounds.py`, evaluates the chain of Section 3 numerically (Remark 3.3 and Table 2). It uses Python 3 and its standard library only. It constructs the six graphs from the extended binary Golay code and the Hoffman–Singleton pentagon-pentagram model, verifies the strongly regular identities $A^2 = kI + \mu(J-I-A)$, verifies that $N = (A-\theta I)(A-kI)$ satisfies $N^2 = (\tau-\theta)(\tau-k)N$

(so that $N/((\tau-\theta)(\tau-k))$ is the orthogonal projection onto the τ-eigenspace, hence positive semidefinite of rank m_τ; this is $M/(k-\tau)$ in the notation of Theorem 1.6), forms the exact rational Gram matrices of all point sets, computes the extension points in the quadratic field of (13), checks every triple, checks the maximality statement of Proposition 7.7, and evaluates the third moments of Remark 7.4. It runs in about ten seconds.

```
#!/usr/bin/env python3
"""Standalone, dependency-free (Python 3 standard library only) verification of the
almost-equidistant sets constructed in the paper.

Every step is exact: integer arithmetic for graphs and idempotents, Fractions for
Gram matrices, and arithmetic in a real quadratic field for the extension points.

A finite set is almost-equidistant if among any three of its points some two are
at distance 1. We certify point sets by their Gram matrices G (entries <p_i,p_j>):
  * realisability in R^d <=> G symmetric PSD of rank <= d (Cholesky/spectral thm),
    which we prove structurally (G is a positive multiple of a symmetric idempotent,
    or a block-diagonal / bordered version thereof);
  * |p_i - p_j|^2 = G_ii + G_jj - 2 G_ij, checked exactly;
  * the almost-equidistant property is checked over all triples.
"""
import itertools, sys
from fractions import Fraction as Fr

# ---------------------------------------------------------------- Golay code
def golay_octads():
    n = 23
    g = [1, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1]       # x^11+x^10+x^6+x^5+x^4+x^2+1
    words = set()
    for m in range(1 << 12):
        cw = [0] * n
        for i in range(12):
            if (m >> i) & 1:
                for j, gj in enumerate(g):
                    if gj: cw[(i + j) % n] ^= 1
        words.add(tuple(cw + [sum(cw) & 1]))
    assert len(words) == 4096
    wd = {}
    for w in words: wd[sum(w)] = wd.get(sum(w), 0) + 1
    assert wd == {0: 1, 8: 759, 12: 2576, 16: 759, 24: 1}
    return [frozenset(i for i in range(24) if w[i]) for w in words if sum(w) == 8]

def steiner_3_6_22():
    blocks = [frozenset(o - {22, 23}) for o in golay_octads() if 22 in o and 23 in o]
    assert len(blocks) == 77
    cnt = {}
    for b in blocks:
        for t in itertools.combinations(sorted(b), 3): cnt[t] = cnt.get(t, 0) + 1
    assert len(cnt) == 1540 and set(cnt.values()) == {1}
    return blocks

# ---------------------------------------------------------------- graphs
def zeros(n): return [[0] * n for _ in range(n)]

def higman_sims(B):
    A = zeros(100)
    for p in range(22): A[0][1 + p] = A[1 + p][0] = 1
    for i, b in enumerate(B):
        for p in b: A[1 + p][23 + i] = A[23 + i][1 + p] = 1
```

```
        for j in range(i + 1, 77):
            if not (b & B[j]): A[23 + i][23 + j] = A[23 + j][23 + i] = 1
    return A

def disjointness_graph(B):
    n = len(B); A = zeros(n)
    for i in range(n):
        for j in range(i + 1, n):
            if not (B[i] & B[j]): A[i][j] = A[j][i] = 1
    return A

def hoffman_singleton():
    A = zeros(50)
    P = lambda h, i: 5 * h + (i % 5); Q = lambda j, i: 25 + 5 * j + (i % 5)
    for h in range(5):
        for i in range(5):
            A[P(h, i)][P(h, i + 1)] = A[P(h, i + 1)][P(h, i)] = 1
            A[Q(h, i)][Q(h, i + 2)] = A[Q(h, i + 2)][Q(h, i)] = 1
            for j in range(5):
                A[P(h, i)][Q(j, h * j + i)] = A[Q(j, h * j + i)][P(h, i)] = 1
    return A

def clebsch():
    A = zeros(16)
    for x in range(16):
        for s in (1, 2, 4, 8, 15): A[x][x ^ s] = 1
    return A

def petersen():
    V = list(itertools.combinations(range(5), 2)); A = zeros(10)
    for i in range(10):
        for j in range(10):
            if i != j and not (set(V[i]) & set(V[j])): A[i][j] = 1
    return A

# ------------------------------------------------------------------ matrices
def matmul(X, Y):
    n = len(X); m = len(Y[0]); k = len(Y)
    return [[sum(X[i][t] * Y[t][j] for t in range(k)) for j in range(m)] for i in range(n)]

def verify_srg(A, v, k, lam, mu):
    n = len(A); assert n == v
    for i in range(n):
        assert A[i][i] == 0 and sum(A[i]) == k
        for j in range(n): assert A[i][j] == A[j][i] in (0, 1)
    A2 = matmul(A, A)
    for i in range(n):
        for j in range(n):
            exp = k if i == j else (lam if A[i][j] else mu)
            assert A2[i][j] == exp
    # triangle-free <=> lam == 0
    assert lam == 0

def srg_spectrum(v, k, lam, mu):
    disc = (lam - mu) ** 2 + 4 * (k - mu); s = int(round(disc ** 0.5)); assert s * s == disc
    theta = ((lam - mu) + s) // 2; tau = ((lam - mu) - s) // 2
    m_theta = ((v - 1) * (-tau) - k) // (theta - tau); m_tau = v - 1 - m_theta
    assert k + m_theta * theta + m_tau * tau == 0
    return theta, tau, m_theta, m_tau
```

```
def tau_projection(A, v, k, lam, mu):
    """Return integer N = (A - theta I)(A - k I) and s with N^2 = s N, so E = N/s is a
    symmetric idempotent (orthogonal projection onto the tau-eigenspace), rank m_tau."""
    theta, tau, m_theta, m_tau = srg_spectrum(v, k, lam, mu)
    n = len(A)
    M1 = [[A[i][j] - (theta if i == j else 0) for j in range(n)] for i in range(n)]
    M2 = [[A[i][j] - (k if i == j else 0) for j in range(n)] for i in range(n)]
    N = matmul(M1, M2); s = (tau - theta) * (tau - k)
    N2 = matmul(N, N)
    for i in range(n):
        for j in range(n): assert N2[i][j] == s * N[i][j] and N[i][j] == N[j][i]
    assert sum(N[i][i] for i in range(n)) == s * m_tau
    return N, s, m_tau, theta, tau

# ------------------------------------------------------------------ point sets
class PointSet:
    """A finite point set given by an exact Gram matrix (list of lists of numbers),
    a proof-of-realisability dimension d, and a description."""
    def __init__(self, G, d, name):
        self.G, self.d, self.name = G, d, name
    def dist2(self, i, j):
        return self.G[i][i] + self.G[j][j] - 2 * self.G[i][j]
    def check_almost_equidistant(self):
        n = len(self.G)
        unit = [[self.dist2(i, j) == 1 for j in range(n)] for i in range(n)]
        for i in range(n):
            for j in range(i + 1, n):
                assert self.dist2(i, j) != 0, "coincident␣points"
        for a, b, c in itertools.combinations(range(n), 3):
            assert unit[a][b] or unit[a][c] or unit[b][c], "bad␣triple␣%s" % ((a, b, c),)
        return n

def srg_point_set(name, A, par):
    """Points P_i = rho E e_i on the sphere |P|^2 = r^2 in the tau-eigenspace (dim m_tau),
    scaled so that non-adjacent pairs are at distance 1. Gram = rho^2 E."""
    verify_srg(A, *par)
    N, s, m_tau, theta, tau = tau_projection(A, *par)
    v = par[0]
    E = [[Fr(N[i][j], s) for j in range(v)] for i in range(v)]
    # non-adjacent pairs have E_ij = e_na; want 2 rho^2 (E_ii - e_na) = 1
    i0, j0 = next((i, j) for i in range(v) for j in range(v) if i != j and A[i][j] == 0)
    rho2 = 1 / (2 * (E[0][0] - E[i0][j0]))
    G = [[rho2 * E[i][j] for j in range(v)] for i in range(v)]
    P = PointSet(G, m_tau, name)
    P.E, P.rho2, P.r2, P.A, P.par, P.tau, P.k = E, rho2, rho2 * E[0][0], A, par, tau, par[1]
    return P

def direct_sum(P, Q):
    """P on sphere |p|^2 = r1^2 in R^d1, Q on sphere |q|^2 = r2^2 in R^d2, r1^2+r2^2 = 1."""
    assert P.r2 + Q.r2 == 1
    n1, n2 = len(P.G), len(Q.G)
    G = [[Fr(0)] * (n1 + n2) for _ in range(n1 + n2)]
    for i in range(n1):
        for j in range(n1): G[i][j] = P.G[i][j]
    for i in range(n2):
        for j in range(n2): G[n1 + i][n1 + j] = Q.G[i][j]
    R = PointSet(G, P.d + Q.d, P.name + "␣(+)␣" + Q.name)
    return R
```

```
def double_simplex(d):
    """Vertices of a regular unit simplex centred at the origin in R^d together with
    their antipodes: 2d+2 points on the sphere |q|^2 = d/(2(d+1))."""
    r2 = Fr(d, 2 * (d + 1)); n = d + 1
    # Gram of S: r2 * ((1+1/d) I - (1/d) J); antipodes: -S
    G = [[Fr(0)] * (2 * n) for _ in range(2 * n)]
    for i in range(2 * n):
        for j in range(2 * n):
            a, b = i % n, j % n
            g = r2 * ((1 + Fr(1, d)) if a == b else 0) - r2 * Fr(1, d)
            if (i < n) != (j < n): g = -g
            G[i][j] = g
    Q = PointSet(G, d, "double␣%d-simplex" % d); Q.r2 = r2
    return Q

def two_cliques(d, c2):
    """Two regular d-point sets of unit vectors with pairwise inner product c2 (<0, 1/|c2|>d),
    scaled to the sphere |q|^2 = r2 = 1/(2(1-c2)); rank d each; placed in the same R^d."""
    r2 = 1 / (2 * (1 - c2))
    # first clique: Gram r2*((1-c2) I + c2 J) on d points (rank d since 1 - c2 + d c2 > 0)
    assert 1 - c2 + d * c2 > 0
    # second clique: the antipodal copy -S (cross inner products -c2 or -1)
    n = d; G = [[Fr(0)] * (2 * n) for _ in range(2 * n)]
    for i in range(2 * n):
        for j in range(2 * n):
            a, b = i % n, j % n
            g = r2 * (1 if a == b else c2)
            if (i < n) != (j < n): g = -g
            G[i][j] = g
    Q = PointSet(G, d, "two␣%d-cliques␣(c=%s)" % (d, c2)); Q.r2 = r2
    # cross distances must not be 1: 2 r2 (1 + c2) != 1 and 4 r2 != 1 (checked by triple test anyway)
    return Q

# ------------------------------------------------ quadratic field Q(sqrt(D))
class Qs:
    """a + b*sqrt(D) with a,b in Q, D a fixed positive non-square rational."""
    D = None
    def __init__(self, a, b=0): self.a, self.b = Fr(a), Fr(b)
    def __add__(s, o): o = o if isinstance(o, Qs) else Qs(o); return Qs(s.a + o.a, s.b + o.b)
    __radd__ = __add__
    def __neg__(s): return Qs(-s.a, -s.b)
    def __sub__(s, o): return s + (-(o if isinstance(o, Qs) else Qs(o)))
    def __rsub__(s, o): return Qs(o) - s
    def __mul__(s, o):
        o = o if isinstance(o, Qs) else Qs(o)
        return Qs(s.a * o.a + s.b * o.b * Qs.D, s.a * o.b + s.b * o.a)
    __rmul__ = __mul__
    def __eq__(s, o): o = o if isinstance(o, Qs) else Qs(o); return s.a == o.a and s.b == o.b
    def __ne__(s, o): return not s.__eq__(o)
    def __hash__(s): return hash((s.a, s.b))
    def sign(s):        # sign of a + b sqrt(D)
        if s.b == 0: return (s.a > 0) - (s.a < 0)
        if s.a == 0: return (s.b > 0) - (s.b < 0)
        t = s.a * s.a - s.b * s.b * Qs.D   # sign(a+b sqrt D) = sign(a) if a^2 > b^2 D else sign(b)
        return ((s.a > 0) - (s.a < 0)) if t > 0 else ((s.b > 0) - (s.b < 0))
    def __repr__(s): return "(%s␣+␣%s*sqrt(%s))" % (s.a, s.b, Qs.D)

def hoffman_cocliques(A, v, k, tau):
```

```
    """All cocliques C with |C| = v(-tau)/(k-tau) such that every vertex outside C has
    exactly -tau neighbours in C (Delsarte/Hoffman-tight cocliques)."""
    s_num, s_den = v * (-tau), k - tau
    if s_num % s_den: return []
    s = s_num // s_den
    nb = [sum(1 << j for j in range(v) if A[i][j]) for i in range(v)]
    out = []
    def rec(chosen, cand, cnt):
        if cnt == s: out.append(chosen); return
        if cnt + bin(cand).count("1") < s: return
        c = cand
        while c:
            i = (c & -c).bit_length() - 1; c &= c - 1
            rec(chosen | (1 << i), c & ~nb[i], cnt + 1)
    rec(0, (1 << v) - 1, 0)
    hoff = []
    for m in out:
        C = [i for i in range(v) if m >> i & 1]
        if all(sum(A[i][j] for j in C) == -tau for i in range(v) if not (m >> i & 1)): hoff.append(C)
    return hoff

def extend(P, cocliques, signs=None):
    """Add one extension point per Hoffman coclique C (with the '+' root).
    x_C = beta * y_C with y_C = 1_U + (k/tau) 1_C, a tau-eigenvector; x_C lies in the span
    of the P_i, so the dimension does not increase. Gram entries:
       <x_C, P_i> = beta*rho*y_i, <x_C, x_C'> = beta^2 <y_C, y_C'>,
    beta determined by |x_C - P_i|^2 = 1 for i in U. Work in Q(sqrt(disc))."""
    v = len(P.G); k, tau = P.k, P.tau
    rho2 = P.rho2; r2 = P.r2
    ys = []
    for C in cocliques:
        y = [Fr(k, tau) if i in C else Fr(1) for i in range(v)]
        # check y is a tau-eigenvector of A (so E y = y):
        for i in range(v):
            assert sum(P.A[i][j] * y[j] for j in range(v)) == tau * y[i]
        ys.append(y)
    y2 = sum(t * t for t in ys[0])
    # beta^2 y2 - 2 beta rho + r2 = 1. Put gamma = beta*rho: (y2/rho2) gamma^2 - 2 gamma + (r2-1) = 0
    # gamma = (1 + sqrt(1 - (y2/rho2)(r2-1))) / (y2/rho2)
    q = y2 / rho2
    Qs.D = 1 - q * (r2 - 1); assert Qs.D > 0
    if signs is None: signs = [1] * len(cocliques)
    gam = {sg: Qs(Fr(1, 1) / q, Fr(sg, 1) / q) for sg in (1, -1)}   # (1 +- sqrt(D))/q
    n = v + len(cocliques)
    G = [[Qs(0)] * n for _ in range(n)]
    for i in range(v):
        for j in range(v): G[i][j] = Qs(P.G[i][j])
    for a, y in enumerate(ys):
        ga = gam[signs[a]]
        for i in range(v): G[v + a][i] = G[i][v + a] = ga * y[i]
        for b, y2v in enumerate(ys):
            gb = gam[signs[b]]
            G[v + a][v + b] = ga * gb * (1 / rho2) * sum(y[i] * y2v[i] for i in range(v))
    R = PointSet(G, P.d, P.name + "␣+␣%d␣extension␣point(s)" % len(cocliques))
    R.qD = Qs.D
    return R

def max_extension_family(P, H):
    """Exact check of the last sentence of Proposition 4.3: the largest family of
```

```
    extension points (from tight cocliques H, both roots) that can be added to P
    inside the span of P while keeping the set almost-equidistant."""
    R = extend(P, H + H, [1] * len(H) + [-1] * len(H))
    v = len(P.G); m = len(R.G) - v
    cocl = [set(C) for C in H + H]
    unit = [[R.dist2(v + a, v + b) == 1 for b in range(m)] for a in range(m)]
    for a in range(m):
        for b in range(a + 1, m): assert R.dist2(v + a, v + b) != 0
    comp = [[a != b and (unit[a][b] or not (cocl[a] & cocl[b])) for b in range(m)] for a in range(m)]
    best = 1 if m else 0
    for a, b in itertools.combinations(range(m), 2):
        if comp[a][b]: best = max(best, 2)
    for a, b, c in itertools.combinations(range(m), 3):
        if comp[a][b] and comp[a][c] and comp[b][c] and (unit[a][b] or unit[a][c] or unit[b][c]):
            best = 3; break
    return best

# ------------------------------------------------------------------ main
def main():
    B = steiner_3_6_22()
    specs = [("Petersen", petersen(), (10, 3, 0, 1)),
             ("Clebsch", clebsch(), (16, 5, 0, 2)),
             ("Hoffman-Singleton", hoffman_singleton(), (50, 7, 0, 1)),
             ("Gewirtz", disjointness_graph([b for b in B if 0 not in b]), (56, 10, 0, 2)),
             ("M22", disjointness_graph(B), (77, 16, 0, 4)),
             ("Higman-Sims", higman_sims(B), (100, 22, 0, 6))]
    sets = {}
    results = []
    def report(P):
        n = P.check_almost_equidistant()
        results.append((P.d, n, P.name)); print("␣f(%d)␣>=␣%d␣:␣%s" % (P.d, n, P.name)); return n
    print("Strongly␣regular␣graphs␣and␣their␣tau-eigenspace␣embeddings:")
    for name, A, par in specs:
        P = srg_point_set(name, A, par); sets[name] = P
        j0 = next(j for j in range(1, len(A)) if A[0][j] == 0)
        print("␣%-18s␣SRG%s␣tau=%d␣m_tau=%d␣r^2=%s␣c_unit=%s"
              % (name, par, P.tau, P.d, P.r2, P.E[0][j0] / P.E[0][0]))
        report(P)
    print("Third␣moments␣sum_{i,j}␣<p_i,p_j>^3␣(exact;␣zero␣iff␣the␣Krein␣condition␣is␣tight):")
    for name in sets:
        G = sets[name].G; v = len(G)
        m3 = sum(G[i][j] ** 3 for i in range(v) for j in range(v))
        print("␣%-18s␣third␣moment␣=␣%s" % (name, m3))
        assert m3 >= 0 and ((m3 == 0) == (name in ("Clebsch", "Higman-Sims")))
    print("Direct␣sums␣with␣a␣double␣simplex␣(partner␣inner␣product␣-1/d2):")
    for name, d2 in [("Clebsch", 3), ("Petersen", 4), ("Hoffman-Singleton", 19),
                     ("Gewirtz", 13), ("M22", 10), ("Higman-Sims", 9)]:
        report(direct_sum(sets[name], double_simplex(d2)))
    print("Higman-Sims␣(+)␣two␣j-cliques␣at␣inner␣product␣-1/9,␣j␣=␣1..8:")
    for j in range(1, 9):
        report(direct_sum(sets["Higman-Sims"], two_cliques(j, Fr(-1, 9))))
    print("Petersen␣in␣its␣theta-eigenspace␣(unit␣pairs␣at␣inner␣product␣-1/3)␣(+)␣Clebsch:")
    A = petersen(); v, k, lam, mu = 10, 3, 0, 1
    theta, tau, mth, mt = srg_spectrum(v, k, lam, mu)
    M1 = [[A[i][j] - (tau if i == j else 0) for j in range(v)] for i in range(v)]
    M2 = [[A[i][j] - (k if i == j else 0) for j in range(v)] for i in range(v)]
    N = matmul(M1, M2); s = (theta - tau) * (theta - k); N2 = matmul(N, N)
    for i in range(v):
        for j in range(v): assert N2[i][j] == s * N[i][j]
```

```
    assert sum(N[i][i] for i in range(v)) == s * mth
    E = [[Fr(N[i][j], s) for j in range(v)] for i in range(v)]
    i0, j0 = next((i, j) for i in range(v) for j in range(v) if i != j and A[i][j] == 0)
    rho2 = 1 / (2 * (E[0][0] - E[i0][j0]))
    Pt = PointSet([[rho2 * E[i][j] for j in range(v)] for i in range(v)], mth,
                  "Petersen (theta-space, r^2=%s)" % (rho2 * E[0][0]))
    Pt.r2 = rho2 * E[0][0]; assert Pt.r2 == Fr(3, 8)
    report(direct_sum(sets["Clebsch"], Pt))
    print("Extension points from Hoffman cocliques:")
    for name, want in [("Petersen", 2), ("Hoffman-Singleton", 2), ("Gewirtz", 2), ("M22", 1)]:
        P = sets[name]; v = len(P.G)
        H = hoffman_cocliques(P.A, v, P.k, P.tau)
        # choose pairwise disjoint cocliques greedily up to 'want'
        chosen = []
        for C in H:
            if all(not (set(C) & set(D)) for D in chosen): chosen.append(C)
            if len(chosen) == want: break
        signs = [1] * len(chosen)
        if len(chosen) < want:      # Petersen: cocliques meet; use the two roots of opposite sign
            chosen = H[:want]; signs = [1, -1]
        print(" %s: %d Hoffman cocliques of size %d; using %d" % (name, len(H), len(H[0]), len(chosen)))
        report(extend(P, chosen, signs))
        if name != "Petersen":
            mx = max_extension_family(P, H)
            print("    largest addable family of extension points (exact, all cocliques, both roots): %d" % mx)
            assert mx == want
    for name in ("Clebsch", "Higman-Sims"):
        P = sets[name]; assert hoffman_cocliques(P.A, len(P.G), P.k, P.tau) == [], name
    print(" (Clebsch and Higman-Sims have no Hoffman cocliques: no extension point exists.)")
    best = {}
    for d, n, nm in results:
        if n > best.get(d, (0,))[0]: best[d] = (n, nm)
    print("\nSummary of certified lower bounds f(d) >= n (published: 2d+4 for d>=10, 24 for d=8,9):")
    for d in sorted(best):
        n, nm = best[d]
        print(" d=%2d n=%3d (2d+4=%3d) %s" % (d, n, 2 * d + 4, nm))
    print("ALL CHECKS PASSED")

if __name__ == "__main__":
    main()
```

References

[1] K. Akiyama, *Nonexistence of a strongly regular graph with parameters* $(266, 45, 0, 9)$*: a certificate-free Lean proof*, preprint (2026), arXiv:2609.08319.

[2] C. Bachoc, B. Bekker, P. Moustrou, F. M. de Oliveira Filho, *Obtuse almost-equiangular sets*, preprint (2025), arXiv:2504.11086.

[3] M. Balko, A. Pór, M. Scheucher, K. Swanepoel, P. Valtr, *Almost-equidistant sets*, Graphs Combin. **36** (2020), 729–754; arXiv:1706.06375.

[4] K. Bezdek, Z. Lángi, *Almost equidistant points on* S^{d-1}, Period. Math. Hungar. **39** (1999), 139–144.

[5] K. Bezdek, M. Naszódi, B. Visy, *On the mth Petty numbers of normed spaces*, in: Discrete Geometry, Monogr. Textbooks Pure Appl. Math. 253, Dekker, New York, 2003, pp. 291–304.

[6] A. V. Bondarenko, *On Borsuk's conjecture for two-distance sets*, Discrete Comput. Geom. **51** (2014), 509–515.

[7] A. E. Brouwer, *The uniqueness of the strongly regular graph on* 77 *points*, J. Graph Theory **7** (1983), 455–461.

[8] A. E. Brouwer, H. Van Maldeghem, *Strongly Regular Graphs*, Encyclopedia of Mathematics and its Applications 182, Cambridge University Press, 2022.

[9] P. J. Cameron, J.-M. Goethals, J. J. Seidel, *Strongly regular graphs having strongly regular subconstituents*, J. Algebra **55** (1978), 257–280.

[10] M. Chen, A. A. Makhnev, M. S. Nirova, *On graphs in which the neighborhoods of vertices are edge-regular graphs without* 3*-claws*, Proc. Steklov Inst. Math. **323** (2023), suppl. 1, S53–S55.
[11] B. Codenotti, P. Pudlák, G. Resta, *Some structural properties of low-rank matrices related to computational complexity*, Theoret. Comput. Sci. **235** (2000), 89–107.
[12] A. L. Gavrilyuk, A. A. Makhnev, *On Krein graphs without triangles*, Dokl. Akad. Nauk **403** (2005), 727–730; English transl. Doklady Math. **72** (2005), 591–594.
[13] A. Gewirtz, *Graphs with maximal even girth*, Canad. J. Math. **21** (1969), 915–934.
[14] C. Godsil, G. Royle, *Algebraic Graph Theory*, Graduate Texts in Mathematics 207, Springer, New York, 2001.
[15] D. G. Higman, C. C. Sims, *A simple group of order* 44,352,000, Math. Z. **105** (1968), 110–113.
[16] A. J. Hoffman, R. R. Singleton, *On Moore graphs with diameters* 2 *and* 3, IBM J. Res. Develop. **4** (1960), 497–504.
[17] T. Jenrich, A. E. Brouwer, *A* 64*-dimensional counterexample to Borsuk's conjecture*, Electron. J. Combin. **21** (2014), no. 4, P4.29.
[18] A. Kupavskii, N. H. Mustafa, K. J. Swanepoel, *Bounding the size of an almost-equidistant set in Euclidean space*, Combin. Probab. Comput. **28** (2019), 280–286.
[19] A. Kovaľ, *The Bannai–Munemasa–Venkov lattice method for strongly regular graphs: a uniform discriminant-form condition and new nonexistence results*, preprint (2026), hexagonmath.org/2610.00225, doi:10.5281/zenodo.23281359.
[20] D. G. Larman, C. A. Rogers, *The realization of distances within sets in Euclidean space*, Mathematika **19** (1972), 1–24.
[21] D. G. Larman, C. A. Rogers, J. J. Seidel, *On two-distance sets in Euclidean space*, Bull. London Math. Soc. **9** (1977), 261–267.
[22] A. A. Makhnev, *The graph* Kre(4) *does not exist*, Doklady Math. **96** (2017), 348–350.
[23] A. Polyanskii, *On almost-equidistant sets*, Linear Algebra Appl. **563** (2019), 220–230.
[24] A. Polyanskii, *On almost-equidistant sets II*, Electron. J. Combin. **26** (2019), no. 2, P2.14.
[25] M. Rosenfeld, *Almost orthogonal lines in* E^d, in: Applied Geometry and Discrete Mathematics, DIMACS Ser. Discrete Math. Theoret. Comput. Sci. 4, Amer. Math. Soc., 1991, pp. 489–492.